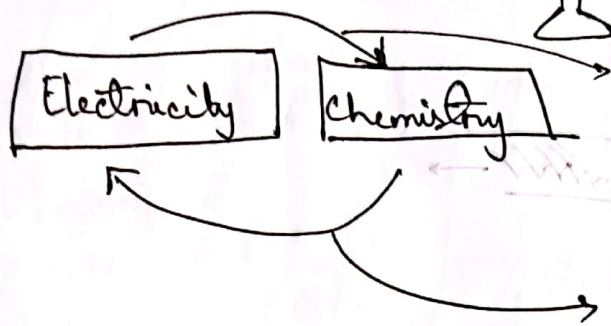


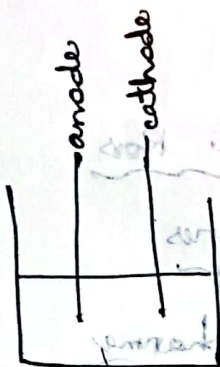
Electrochemistry



using electricity to make chemical reaction (Electroplating)

using chemical reaction to produce electricity

(Phone Batteries)

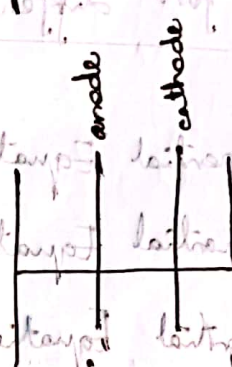


electrolytic cell

endothermic

takes up energy

anode: oxidation



galvanic cell

exothermic

gives off energy

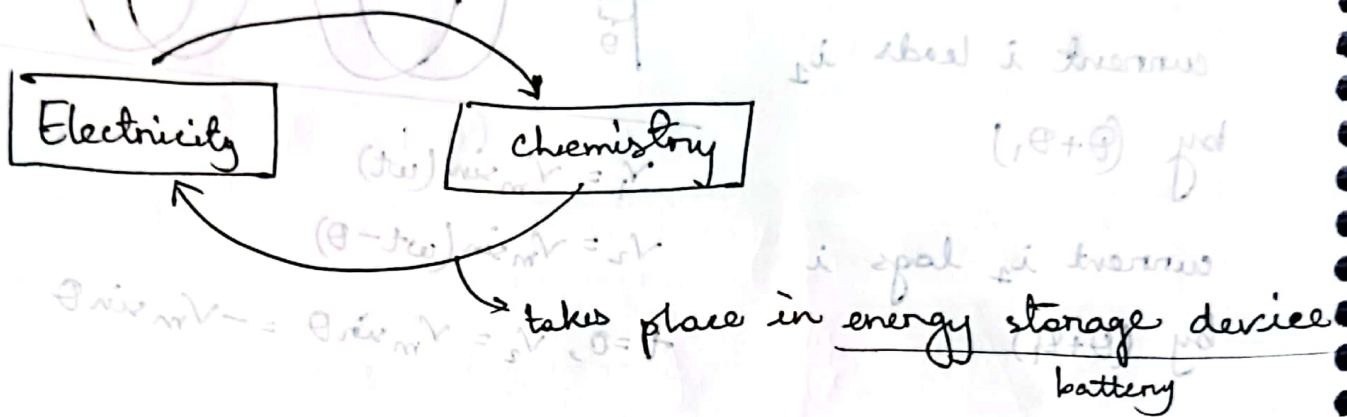
☆ Electrochemistry

Higher Engineering Mathematics of E & C

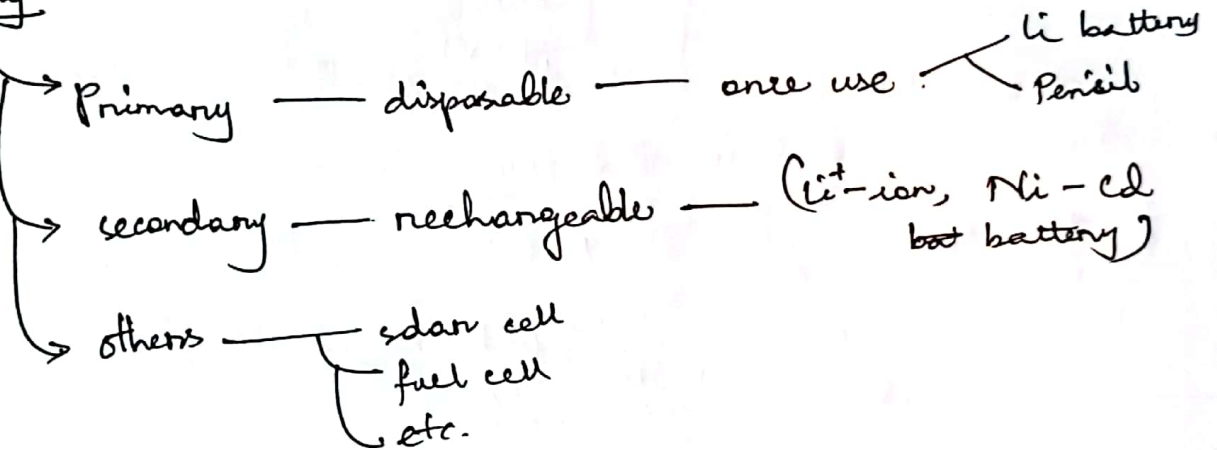
1. Electrolytic cell endothermic oxidation

2. Galvanic cell exothermic reduction

Electrochemistry



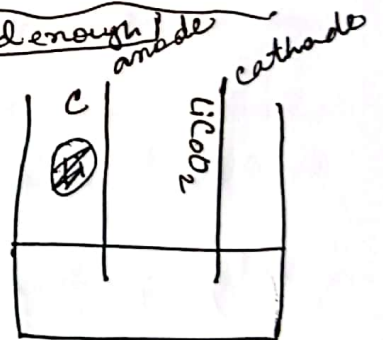
Battery



Lithium Battery:

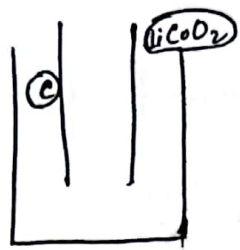
developed by: John Goodenough

(Li) is highly reactive, so making equipments and things with it makes the equipments explosive.



But if (Li) is encapsulated inside LiCoO_2

LiCoO₂, the reactivity reduces, but we get the full power.



electrolytes:

LiFePO_4 — organic

Li^+ battery — non-aqueous battery

(9.11) (10.11) January 10

SEI layer

→ Solid Electrolyte Interface

$$0 = \mu + \frac{\partial \mu}{\partial x}$$

(9.11) (10.11) January 10

A problem involving one or more of the following conditions is related to the value of β and the at details conditions in the following

$$0 = (N) \mu \quad ; \quad 0 = \mu + \frac{\partial \mu}{\partial x}$$

$$Z = (X) \mu$$

of the following conditions

(9.11) (10.11) January 10

different values of independent

of the following conditions

$$0 = (X) \mu \quad ; \quad 0 = \mu + \frac{\partial \mu}{\partial x}$$

$$Z = \left(\frac{X}{Z}\right) \mu$$

of the following conditions

- * Lithium ion battery
- * Li polymer battery
- * SEI layer
- * Units

difference:

Li-ion and LP have the same anode and cathode, but electrolytes are different.

Li-ion
→ LiPF_6 in ether

Li-Polymer
→ Polyethylene / Polyacrylonitrile

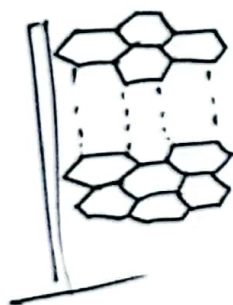
Electrolyte:

→ LiPF_6 in ether

→ Polyethylene oxide or Polyacrylonitrile

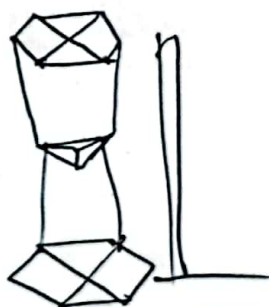
Anode:

Graphite:



Cathode

LiCOO_2



□ Why we don't use Li-ion battery in mobile batteries?

→ LiPF_6 in ether is not solid, ether is a kind of solvent. This is why it is called Gel battery. Using it in mobile might cause it to leak out.

But Li-Polymer batteries are made of solid. They won't leak.

LP $\rightarrow$ Li-Polymer
850 mAh $\rightarrow$ 850 mA current can be supplied for 1 hour
3.7 V $\rightarrow$ voltage difference between anode and cathode is 3.7 V.

Li-ion reaction mechanism:

positive electrode: $\text{LiCoO}_2 \xrightleftharpoons[\text{discharging}]{\text{charging}} \text{Li}_{1-x}\text{CoO}_2 + x\text{Li}^+ + xe^-$
Through electrolyte

negative electrode: $x\text{Li}^+ + xe^- + 6\text{C} \xrightleftharpoons[\text{discharging}]{\text{charging}} \text{Li}_x\text{C}_6$
Through lead

overall: $\text{LiCoO}_2 + \text{C}_6 \xrightleftharpoons[\text{discharging}]{\text{charging}} \text{Li}_{1-x}\text{CoO}_2 + \text{C}_6\text{Li}_x$

SEI layer: Solid Electrolyte Interface layer

* Why SEI layer is negative?

* Reaction mechanism of SEI layer.

* How SEI layer forms?

$\rightarrow$ when lithium⁺ ion moves through electrolyte, it attracts the neg(-ve) part of electrolyte, and when it goes into the other side, the SEI layer is formed and it is negative.

CHEM-1207
m.iQUESTIONS discussed
before in previous classes

- ☐ Write down the reaction mechanism of Li^+ battery with schematic diagram.
- ☐ What are the differences between Li^+ battery and Li-polymer battery?
- ☐ What are the advantages/disadvantages of Li^+ battery?
- ☐ Classifications of battery/units of battery.
- ☐ What is SEI layer? Why is it negative in charge?
- ☐ How SEI layer forms?

Ni-Cd battery:



spontaneous redox
Exothermic



unstable form
high energy

stable form
low energy

releases energy as heat

but when we use battery, we want to use this heat to generate electricity instead

① Ni-Cd battery
② Fundamentals of electrochemistry
③ EMF / Voltage

Q What is the difference between EMF and voltage?

→ EMF = the force needed to make an electron go through the whole circuit

Voltage = difference of volts between two points

Ni-Cd battery:



→ rectangular



→ cylindrical



→ circular

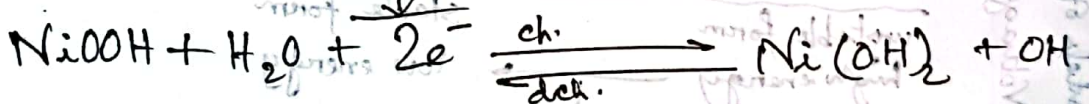
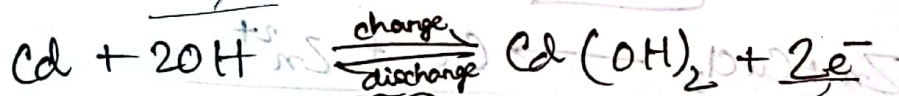


→ pouch
can be flexible

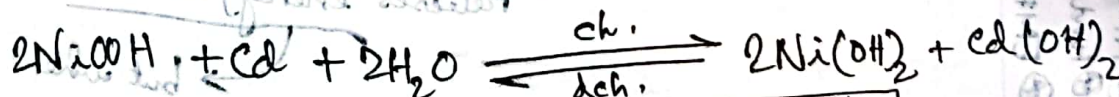
reaction mechanism:

anode (-ve)

cathode (+ve)



overall:



Nickel oxyhydroxide

Ni is in here like NiOOH, valency 3+, does red

Cd is in here like Cd, valency 0, does ox

secondary battery : those who can be recharged

calculator, transistor made by Ni-Cd

Ni-Cd

* Double layer capacitance :

Difference
between
battery &
capacitor

→ Battery holds charge and discharges slowly,
capacitor discharges instantly

→ Battery converts chemicals to electricity, whereas
capacitor ^{provides} ~~converts~~ electricity charge ~~to charge~~

Battery: Chemical → charge
Capacitor: Charge → Charge

→ Capacitor works quick because there is no conversion, it just stores charge inside it.

CHEM 1203
battery

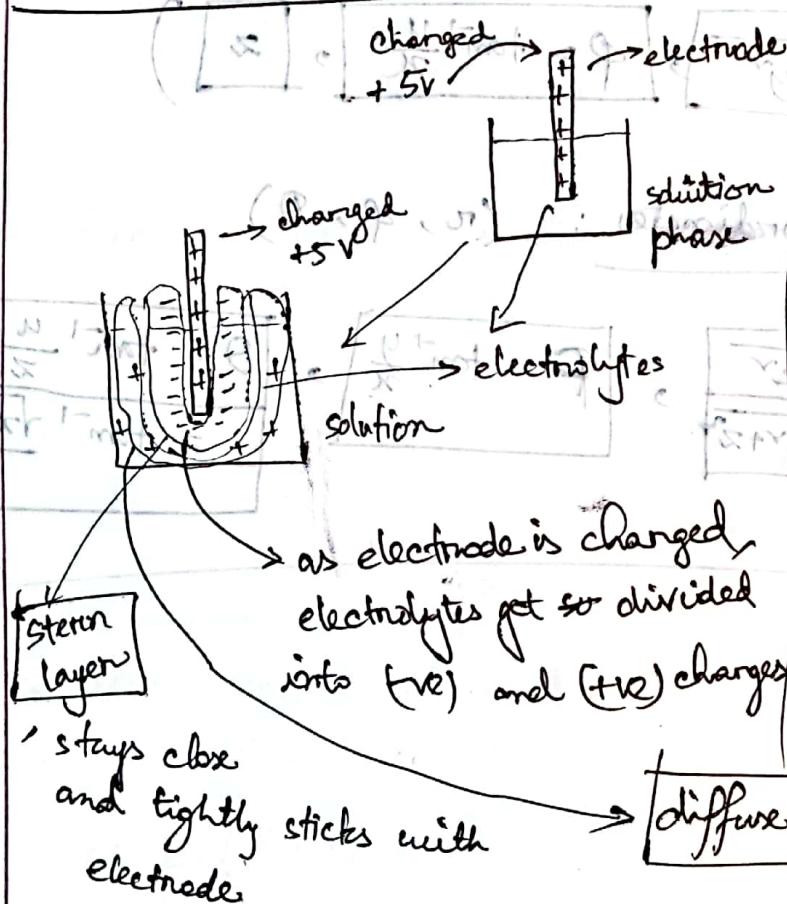
today's class:

* Double layer capacitance

* Zeta potential

* Nernst equation

* Ni-Cd battery

Double Layer Capacitance

in Ni-cd battery

 $\text{Cd} \rightarrow \text{Cd}^{2+}$ (anode) $\text{NiOOH} \rightarrow \text{Ni}^{2+}$ (cathode)

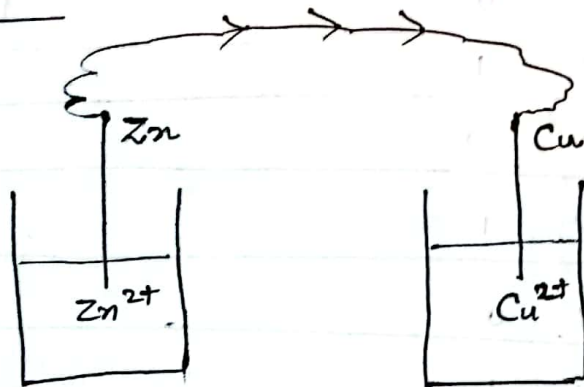
Ni turns to NiO ,
and then NiO
turns to NiOOH

 $\text{Cd} \rightarrow \text{Cd}(\text{OH})_2$ $\text{NiOOH} \rightarrow \text{Ni}(\text{OH})_2$

Zeta Potential :

Potential difference between 'stern layer' and 'diffuse layer'.

Nernst equation :



electric work

$$Q \times \text{EMF}_{\text{cell}} = Q E_{\text{cell}} = n F E_{\text{cell}} = -n F E$$

$$1 e^- = 1.602 \times 10^{-19} \text{ C}$$

$$\begin{aligned} 1 \text{ mole } e^- &= 6.02 \times 10^{23} \times 1.602 \times 10^{-19} \text{ C} \\ &= 96500 \text{ C} \\ &= 1 F \end{aligned}$$

$$n \text{ mole} = n F$$

$$\text{Total work done} = -\Delta G$$

according to thermodynamics,

$$\Delta G = \Delta G^\circ + RT \ln Q$$

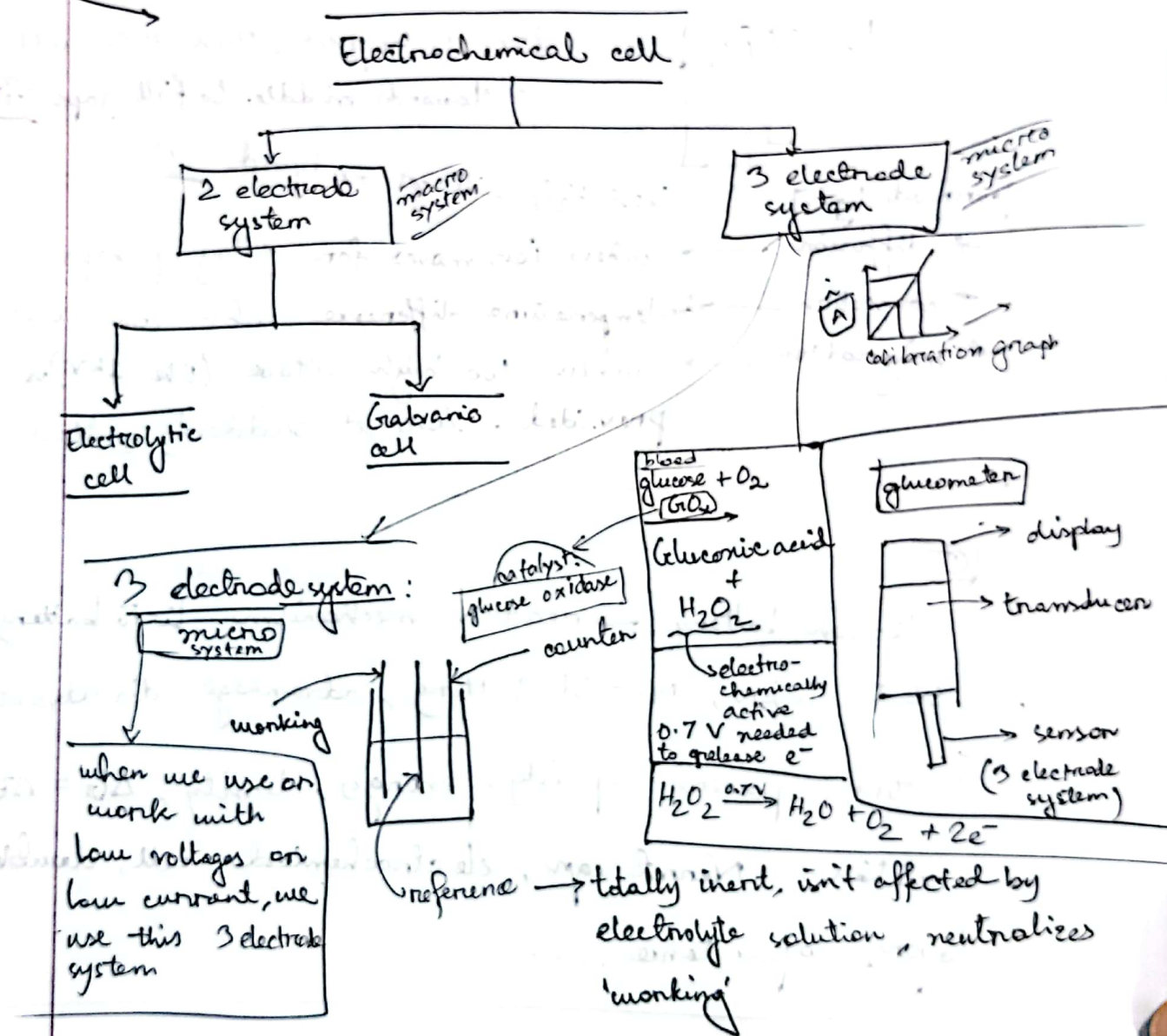
$$-n F E = -n F E^\circ + RT \ln Q$$

$$\Rightarrow E_{\text{cell}} = E^\circ - \frac{RT}{nF} \ln Q$$

$$\Delta G = -n F E$$

(-) because it gives us energy

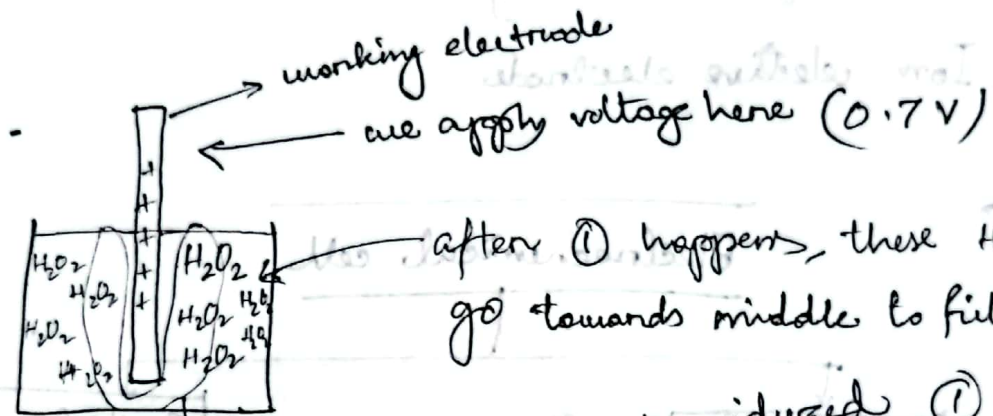
- * Electrochemical cell
- * Mode of mass transfer
- * Ion selective electrode



Polyacrylonitrile

[Aboltee] → sir worked here

Mode of mass transfer :



current types

- * diffusion → when ion moves for filling up gaps
- * convection → for temperature difference, when ion moves
- * migration → when too high voltage (like 1kV) is provided, ions get suddenly pulled

[CT]

Li-ion battery → reaction mechanism, Li-Pb battery, SEI layer, Ni-Cd battery, advantage, disadvantage, battery, power, capacity, energy, density, $\Delta G = \Delta E$ relation, Nernst eqn, electrochemical cell, double layer, capacitance, ...

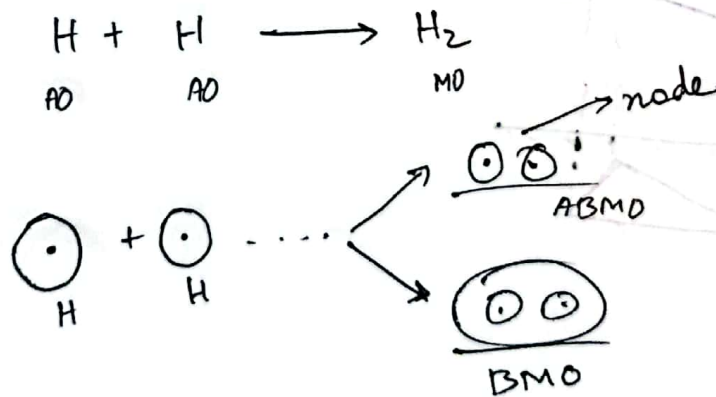
CHEM 1207
MOMChemical Bonding:

- * Ionic
- * Covalent
- * Metallic
- * Hydrogen bond
- * Vander Wall's theorem

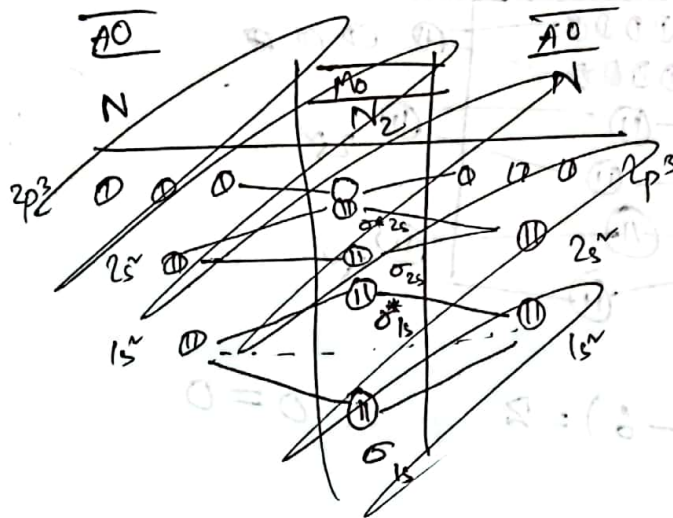
Modern Inorganic
Chemistry
SZ Haider

⇒ Covalent:

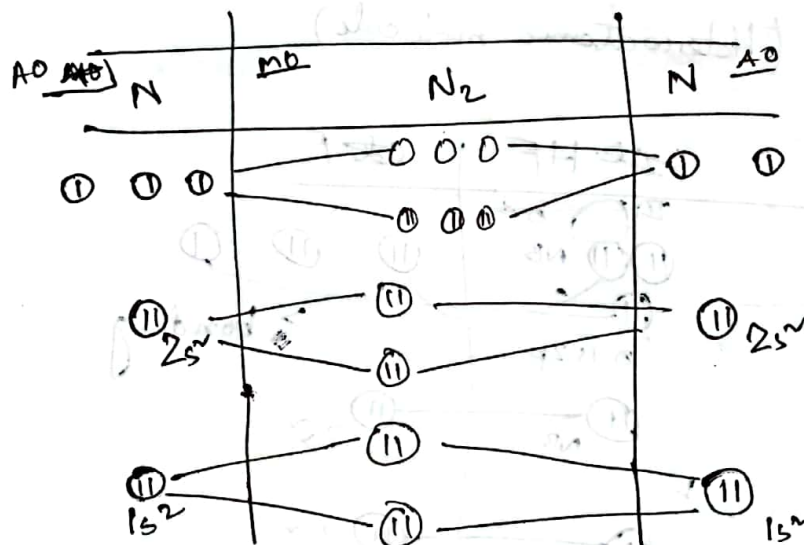
- Valence bond theory → [can't explain well]
- Molecular orbital theory

Molecular Orbital Theory:

7 Draw a molecular orbital diagram of N_2 molecule & mention it.



$\sigma^* =$ ~~Not~~ Antibonding



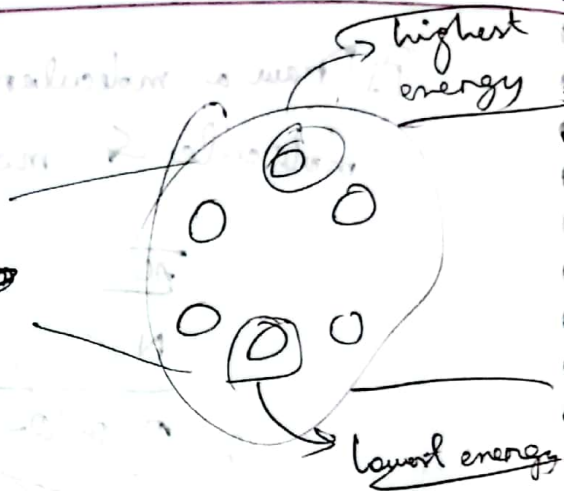
$$\therefore \frac{1}{2} \text{ Bond order} = \frac{1}{2} (10 - 4) = 3$$

$\therefore N \equiv N$

$\Rightarrow$ has single e^- in

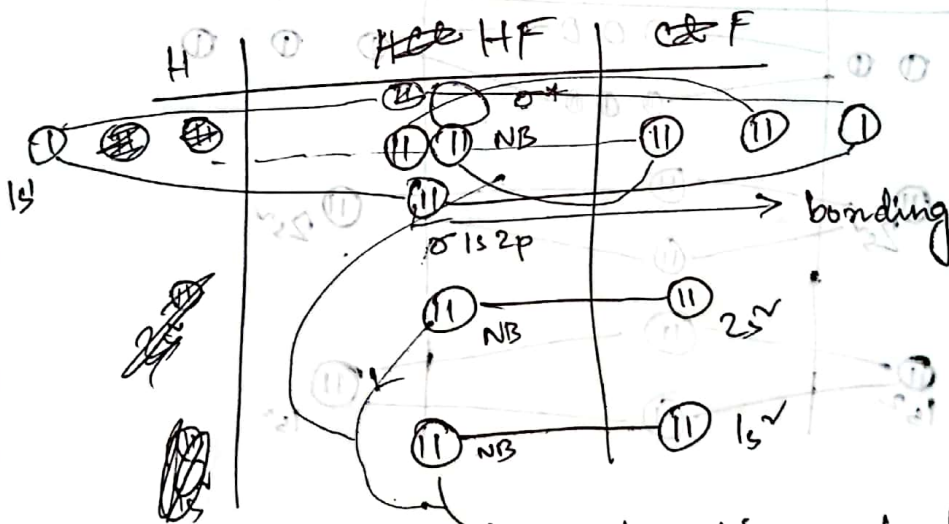
$\rightarrow$ this means there will be 3 bonds, between $N \equiv N$, in N_2 .

O_2 more pairs involved solidus involved in reaction



$$0 = 0$$

☒ HF (Heteroatomic molecule)



on bonding molecular orbital,
energy level same as before

$$B.O = \frac{1}{2}(R)$$

Chemical bonding

↓
* VBT

↘ hybridization
↘ resonance

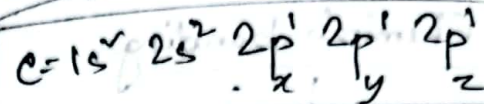
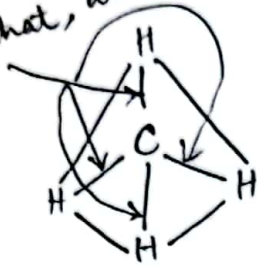
* metallic bond

* H-bond

Hybridization

no of orbitals	hybrid orbitals	geometry	example	structure
2	sp	linear	$BeCl_2$	
3	sp^2	triangular	BBr_3	
4	sp^3 / dsp^2	tetrahedral / square planar	CH_4	
4	dsp^2 / $sp^2 d$	square planar	$Ni(CN)_4^{2-}$	
5	$sp^3 d$ / dsp^3	trigonal bipyramidal	PCl_5	
6	$sp^3 d^2$ / $d^2 sp^3$	octahedral	SF_6	
7	$d^3 sp^3$	pentagonal bipyramidal		

(XRD) proves that, all these distances are same. $\rightarrow$ but these orbitals aren't same energy

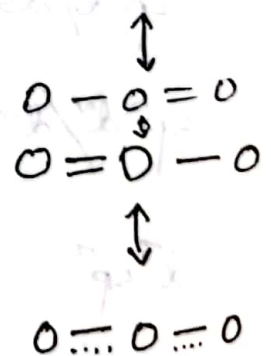
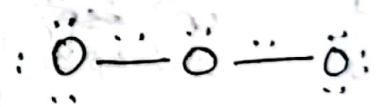
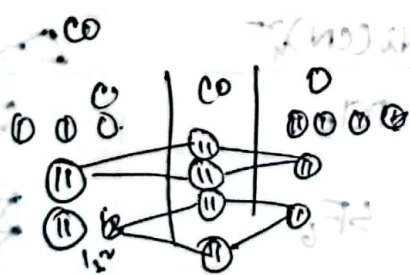
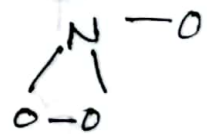
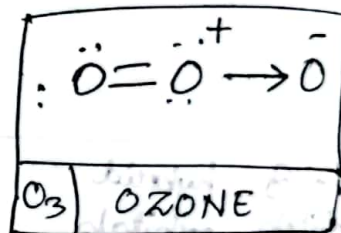
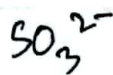
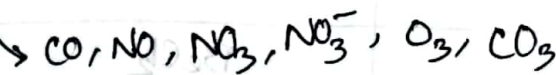


* central atom determines the structure.

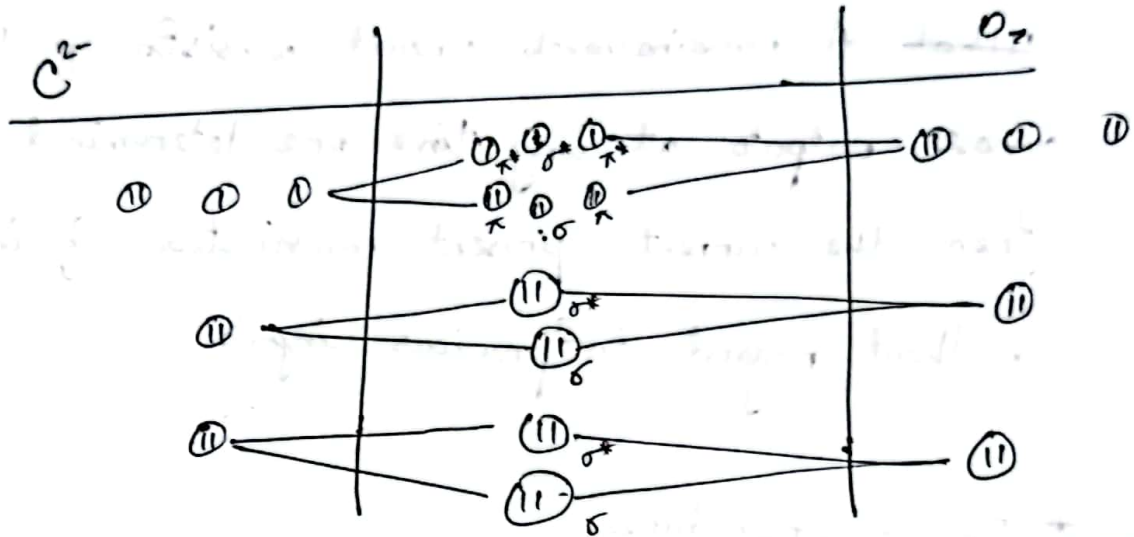
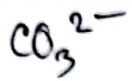
Describe hybridization of SF_6 and draw its structure.

Resonance structure:

problematic structures



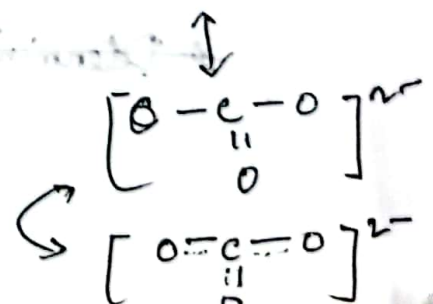
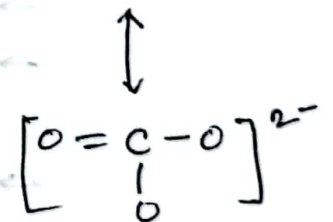
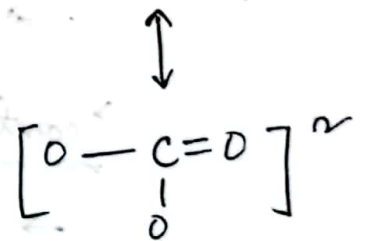
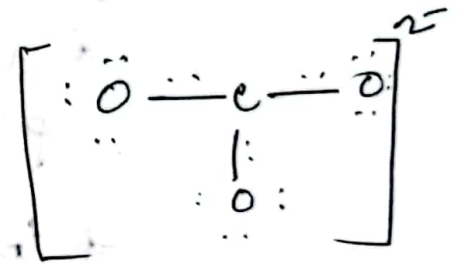
O_3 resonance structure

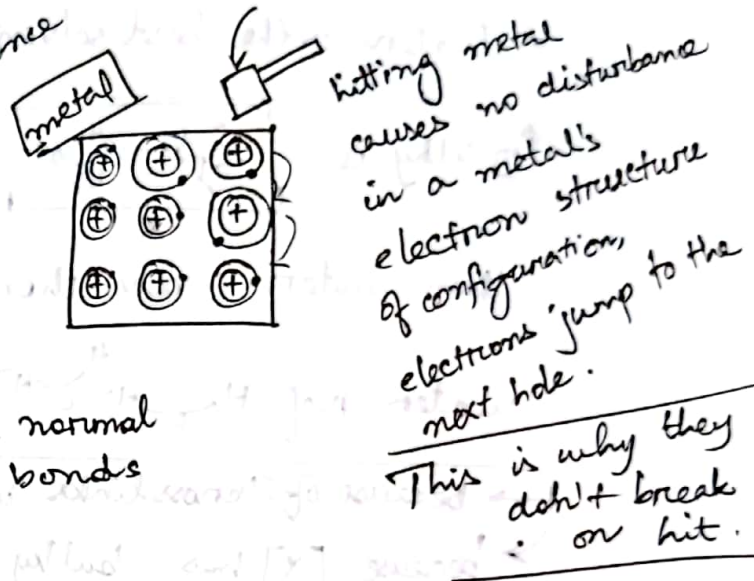
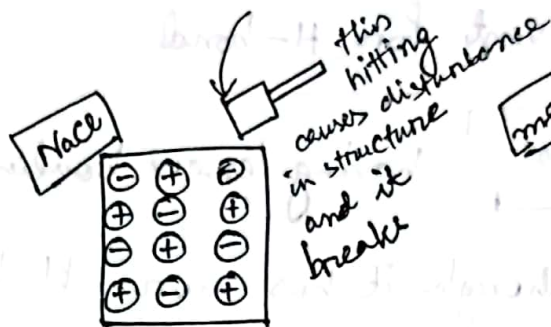


$$BO = \frac{1}{2}(10 - 6) = 2$$

Metallic bond :

electrostatic attraction between metal and electron.



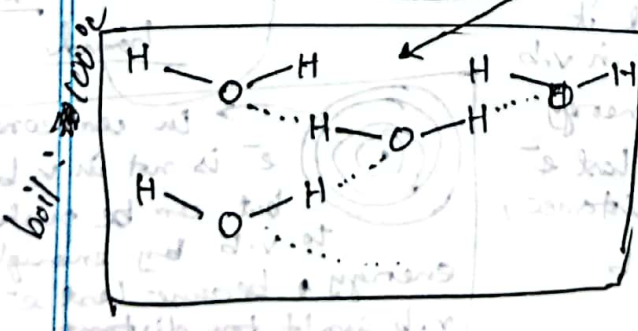
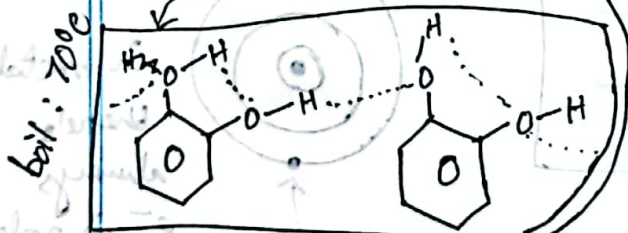


A metal is normally solid in normal conditions, while covalent bonds creates

temperature increase causes disruption in metal's e^- structure and configuration

H-bond:

↳ intramolecular
↳ intermolecular



Questions:

(1) Metals are malleable & ductile, explain with diagram.

(2) With increase in temperature conductivity of metals decreases, explain.

(3) Explain inter & intra molecular H-bond.

(4) Draw the crystal structure of Si & mention the number of atoms present in a unit cell of Si.

metallic bond
H-bond
Si crystal structure

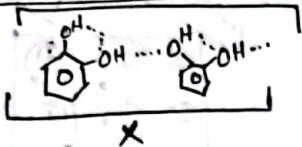
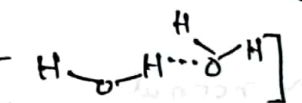
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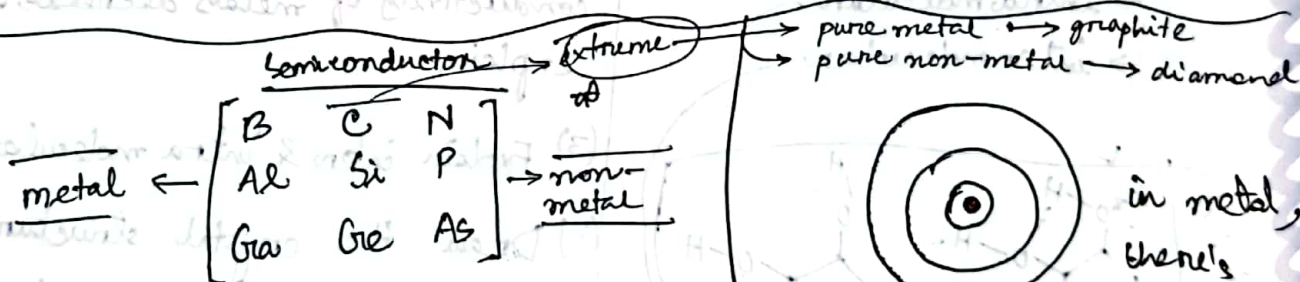
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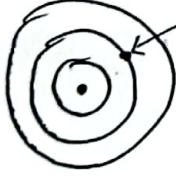
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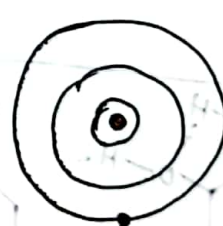
* Water is the best solvent for H-bond

* Why is  having less boiling point than water even though it has more H-bonds than water H_2O  ?

→ because of crosslinks in $[X]$'s $[O \cdots H]$
 → because $[X]$ has bulky  compound part which doesn't form any H-bond, so we can only apply energy to $[OH \cdots OH]$ parts and it will get ^{disassociated} dissolved.
 But in H_2O , we have to apply energy in both H and O.



 in non-metal, there's no e^- in v.b, and it can never have an e^- in v.b even with ∞ amount of energy given to it, because the last e^- is in such state and distance, that no energy can push it to v.b

 in metal, there's always e^- in valence band

 in semiconductor, e^- is not in v.b, but can be pushed to v.b by enough energy, because last e^- and v.b isn't too distant

Topic Name : _____

Day : _____

Time : _____

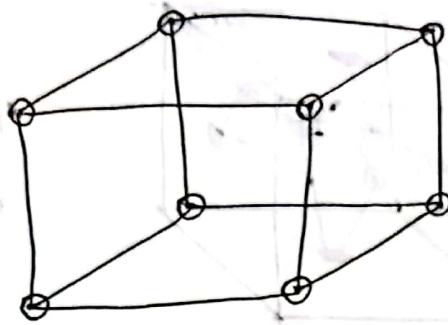
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* C is special because it's extreme, it can be in both pure metal (Graphite) and both pure non-metal form (Diamond).

* Si is used more in semiconductors, because it is cost effective.

* 3-5 LAB

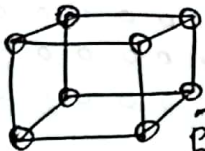
* Silicon Carbide has good fluctuation



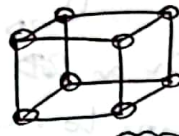
* Si / Ge crystal structure

* Silicon's crystal structure is diamond like cubic structure.

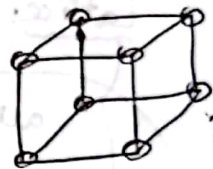
cubic



primitive



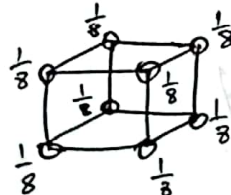
FCC



BCC

* cubic crystals can be of 3 types.

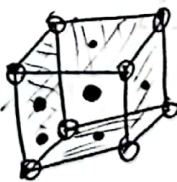
↳ primitive:



there is 8 atoms in here

$$8 \times \frac{1}{8} = 1$$

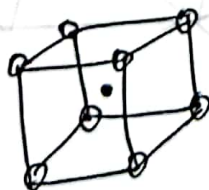
↳ FCC (Face centered cubic):



$$6 \times \frac{1}{2} = 3$$

there is 3 in here

↳ BCC (Body centered cubic):



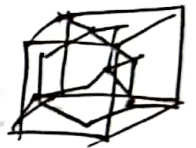
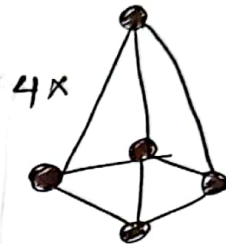
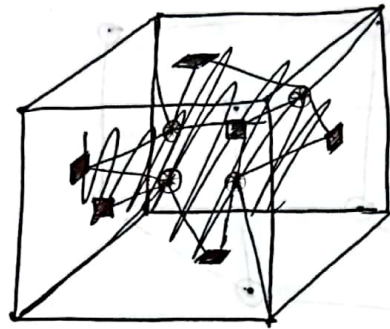
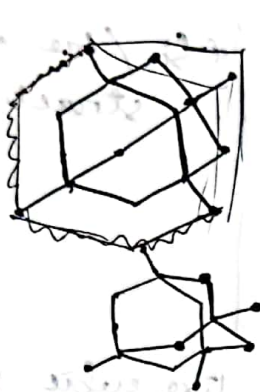
2 in here.

Topic Name : _____

Day : _____

Time : _____

Date : / /



Lattice + Bravais = crystal

all the points in a ^{3D} plane must be similar to each other

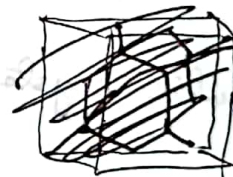
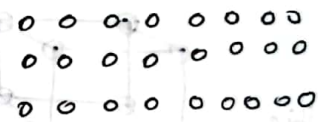
points distribution in a regular pattern is

defined atoms/ions/molecules having homogeneous distribution

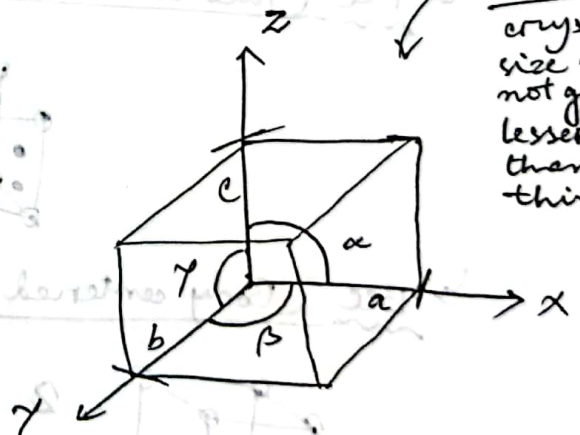
there are no pure crystals

7 crystals:

- ↳ cubic
- ↳ trigonals
- ↳ hexagonal
- ↳ orthorhombic
- ↳ tetragonal
- ↳ monoclinic
- ↳ rhombohedron



unit cell:
crystal size can not go lesser than this

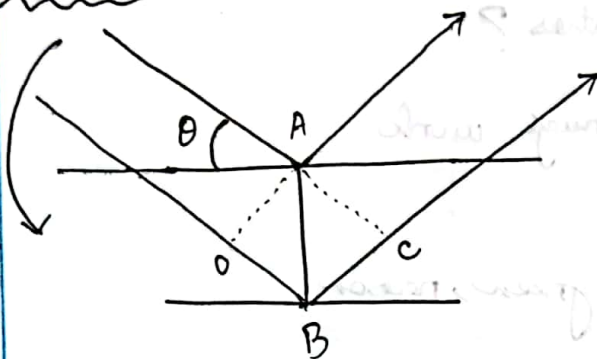


Topic Name: CHEM 1207 (Crystal)

Day: Monday

Time: 11:30 Date: 11/09/23

X-Ray:



$$\begin{aligned} n\lambda &= OB + BC \\ &= 2AB \sin \theta \\ n\lambda &= 2d \sin \theta \end{aligned}$$

$$\begin{aligned} OB &= AB \sin \theta \\ OB &= BC \end{aligned}$$

→ Bragg's equation
→ XRD formula

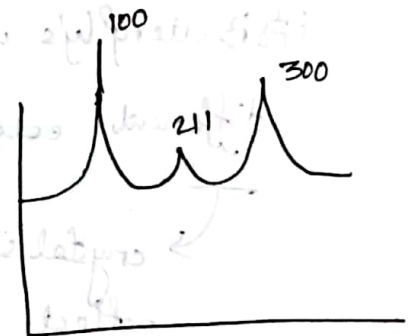
* atom's size $\approx$ wavelength of X-Ray
that's why we use

* upon rotating the θ

Solids

→ crystalline: salt, sugar

→ amorphous: rubber, plastics, glasses



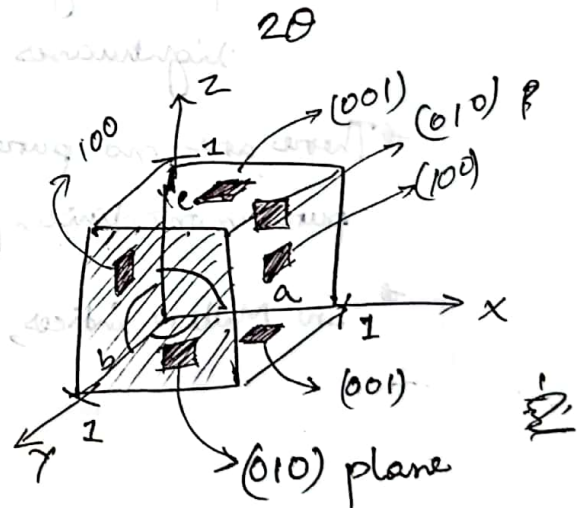
Miller indices:

intercepts: $\infty, 1, \infty$

reciprocal: $\frac{1}{\infty}, \frac{1}{1}, \frac{1}{\infty}$

$= 0, 1, 0$

$\therefore$ Miller indices = (010)



Q Why we need crystal properties ?

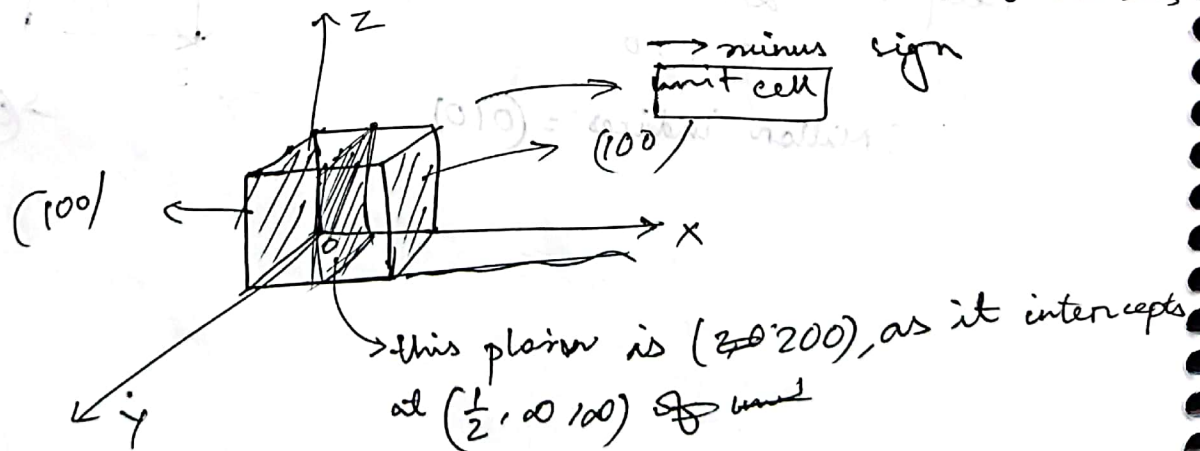
- for discovering how drugs work
- why diamonds shiny
- rubies red, emeralds green, reason.
- used in nanotech

Q Butterfly's wings don't have so much pigment, but it shows different colours, multiple colours, why?

- crystal's size difference, different plane of the crystals reflect light differently, size's difference makes the lightwaves different

* There are no pure crystal structures, (no pure triclinic, pure monoclinic, pure orthorhombic)

* In Miller's indices, there can be no → fraction [handled by turning into int]



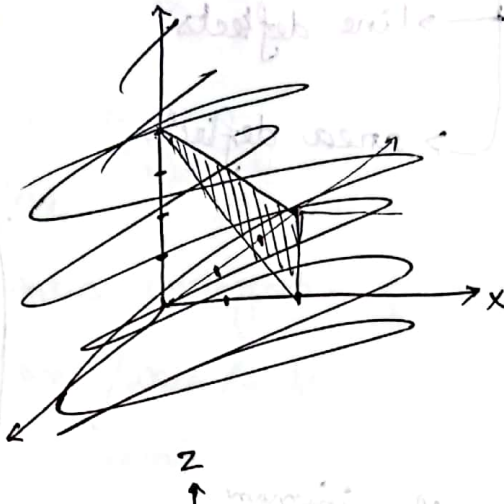
Topic Name : _____

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☐ $2a, -3b, 4c$



Miller indices

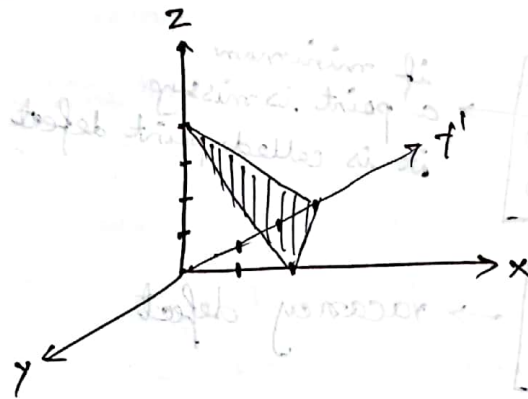
intercepts : $2, -3, 4$

reciprocal : $\frac{1}{2}, -\frac{1}{3}, \frac{1}{4}$

$\Rightarrow \frac{6}{12}, -\frac{4}{12}, \frac{3}{12}$

$\Rightarrow (6, \bar{4}, 3)$

$\Rightarrow (6\bar{4}3)$



imperfection in Solids :

☐ XRD $\rightarrow$ xray of crystals : explanation



square

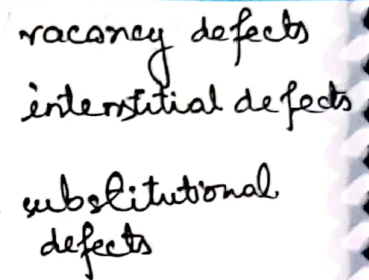
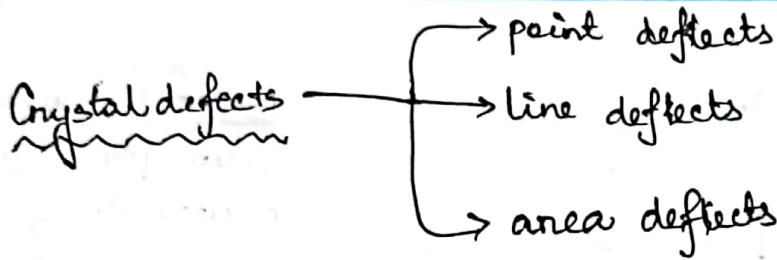


square

square

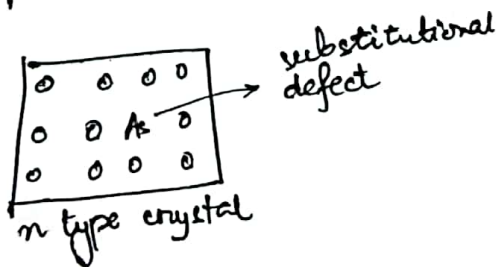
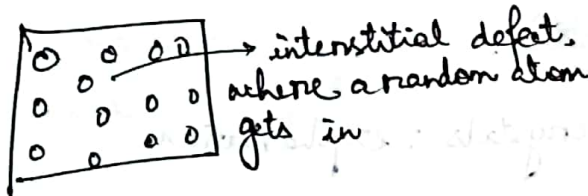
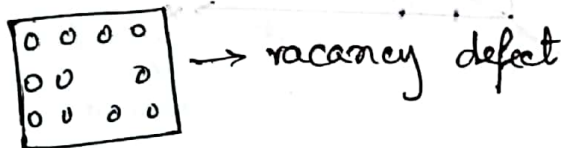
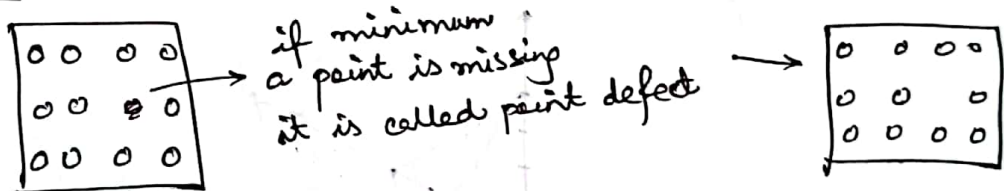


square

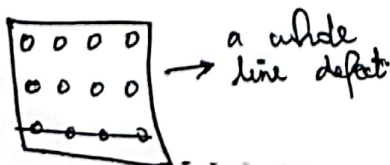


Q: Explain various crystal defects with diagram

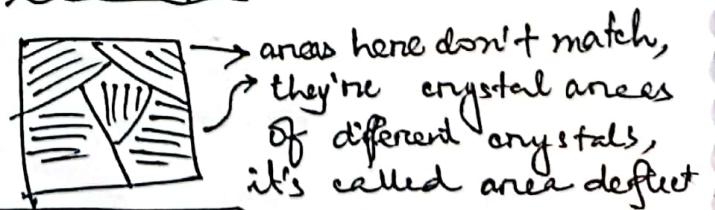
Point defects:



line defects:



area defect



Polycrystalline

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~~A~~ crystal defect is a desirable (mostly) crystal property.

Q.A

→ Crystal : periodic arrangement of molecules/ions/atoms in a space

→ Lattice : periodic arrangement of points in a space

→ Bravais : atoms/ions/molecules

→ Miller indices : naming of planes

→ XRD : use 10^{-10} nm wavelength because distance of a unit cell's 2 points is close to 10^{-10} nm

↳ Bragg's theory (MATH Problems)

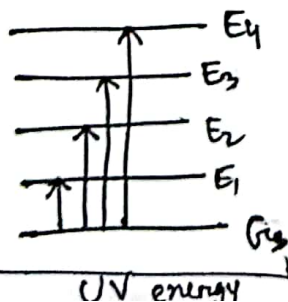
→ Crystal defects : crystals are always having defects, for defects they can be molded, modified. Pure crystals are very rare.

SPECTROSCOPY



the way we know that H_2O molecules are like this is by XRD.

it is always in motion, and that motion generates energy



UV energy



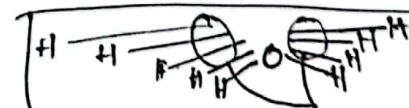
UV ray



IR



microwave



stretching of H_2O

IR

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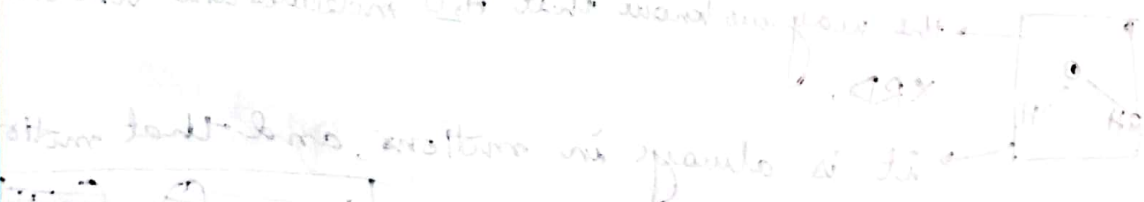


these bonds rotate, and it needs microwave ray energy

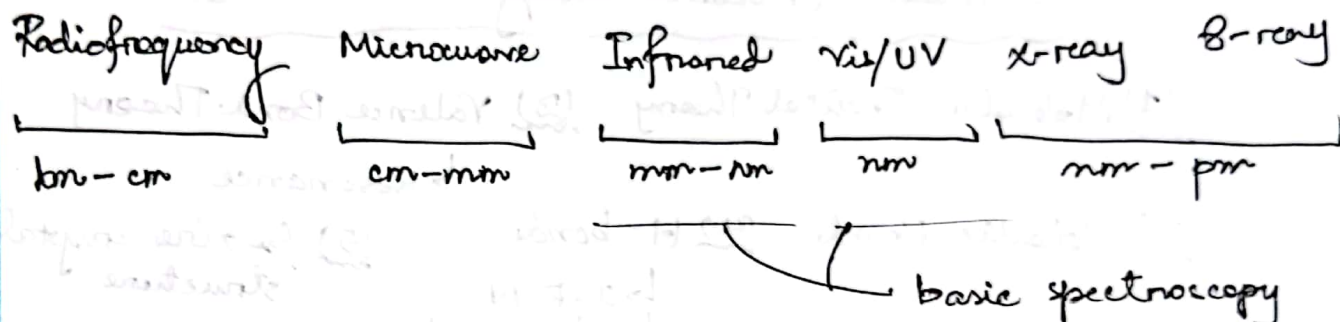
energy state transitioning happens only with UV ray energy

stretching & vibration of atoms in molecule happens for IR ray energy

SPECTROSCOPY



Electromagnetic Spectrum

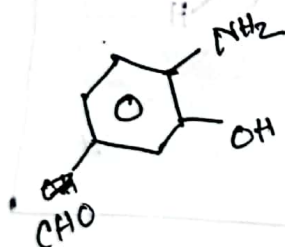


Spectroscopy :

electromagnetic radiation and matter's interaction

☐ Why infrared is needed:

↳ determining organic molecule (be it crystalline or not) and also be able to quantify it & detect all its functional groups



↳ why? : Because the vibration energy of the molecular groups match with IR's energy. Resonance ≠ vibration

① ☐ Explain the instrumentation of IR spectroscopy.

② ☐ Explain the different vibration mode of molecules with diagrams.

Topic Name : _____

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Date : / /

Lecture based on C.T (Chemical Bonding)

S. Z. Haider (Modern Inorganic Chemistry)

(1) Molecular Orbital Theory (2) Valence Bond Theory

↳ Resonance

(3) Metallic Bond

(4) H-bonds

(5) Si & Ge crystal structure

(Boltz Tuli)
Phy chem

↳ O, F, N

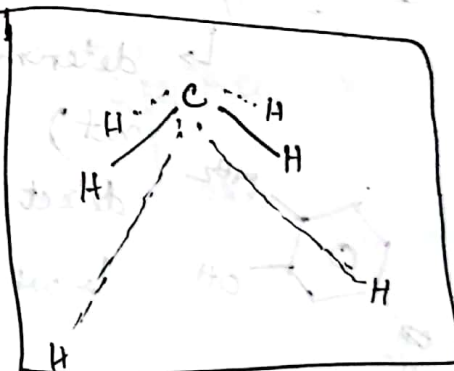
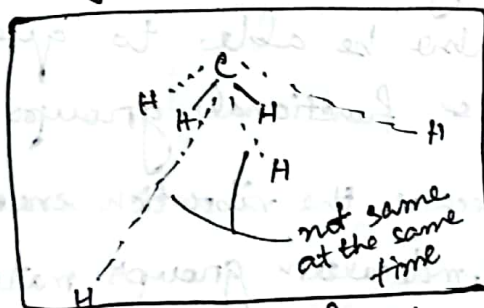
↳ melting point

↳ boiling point

↳ intra, inter

④ Different vibrational mode of a molecule:

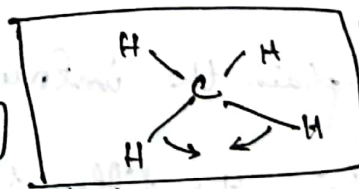
↳ symmetric stretching :



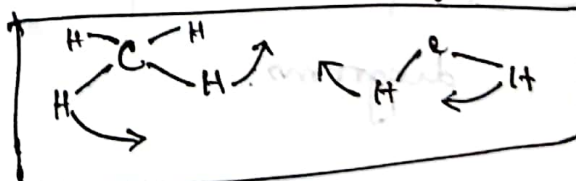
↳ asymmetric stretching :

↳ Bending

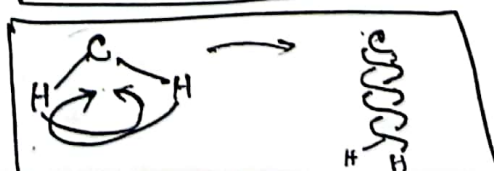
↳ Scissoring



↳ Rocking



↳ Twisting



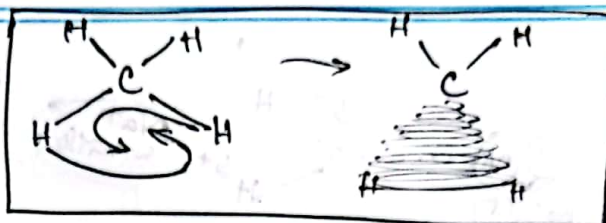
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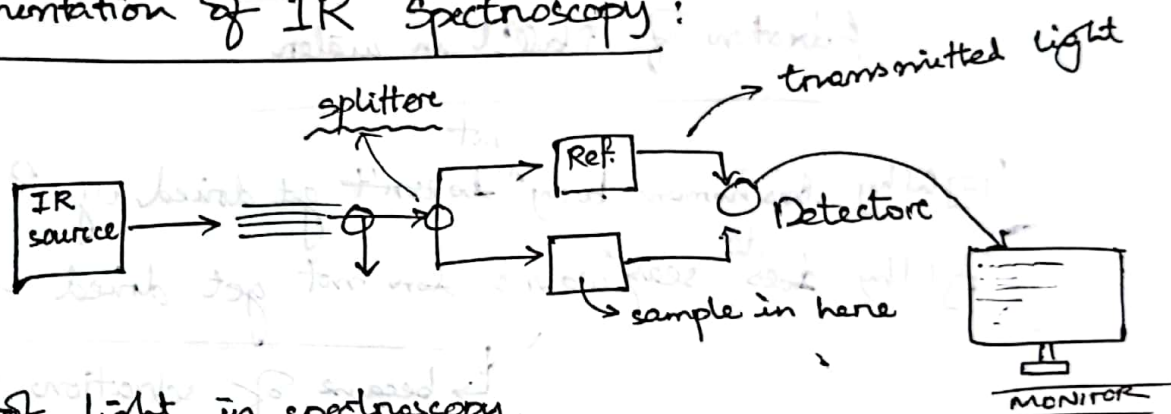
Date : / /

↳ Wagging :

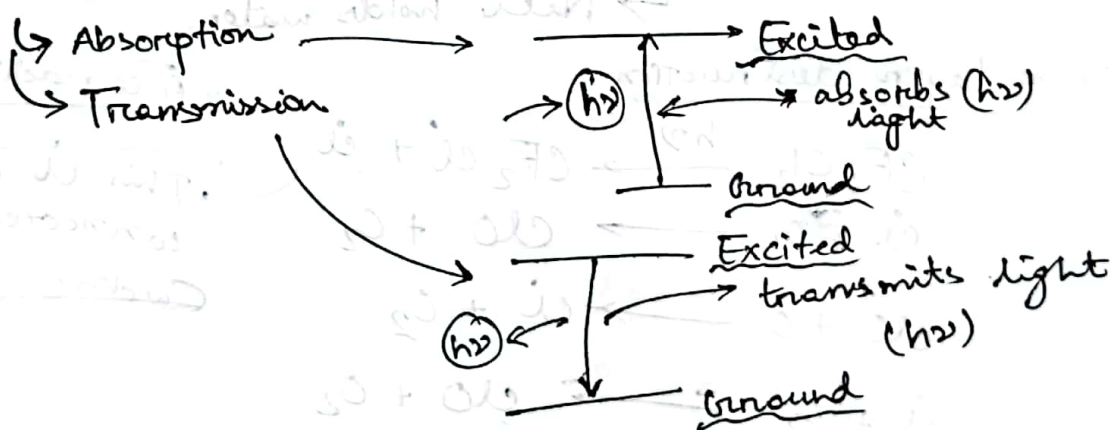


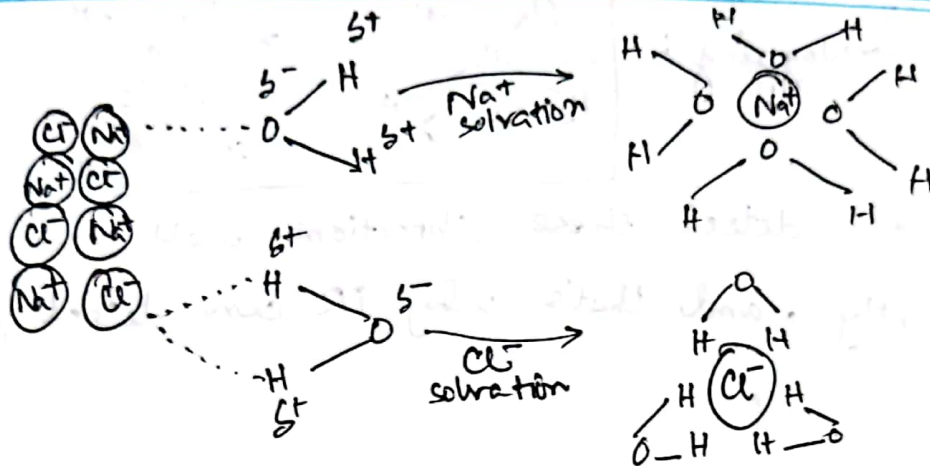
* IR can detect these vibrations well and all of them differently, and that's why IR can detect groups.

⑧ Instrumentation of IR Spectroscopy :



2 types of light in spectroscopy.





Solvation of NaCl in water

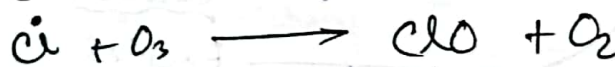
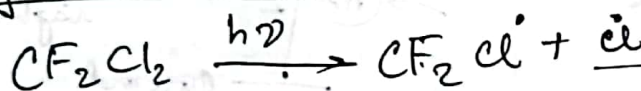
□ Why does human body ^{not} ~~doesn't~~ get dried up?

□ Why does ^{do} seas/oceans ~~don't~~ get dried up?

↳ because of solvation of NaCl in water

↳ NaCl holds water.

Ozone layer destruction:



cycle

→ This Cl breaks many O₃ molecules

Free radical

This Cl was connected with Carbon

Topic Name : _____

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Composition:

Troposphere:

Nitrogen $\rightarrow$ 78%

Oxygen $\rightarrow$ 21%

Others $\rightarrow$ 1%

Stratosphere:

[same as Troposphere]

$\rightarrow$ Stratosphere

Mesosphere:

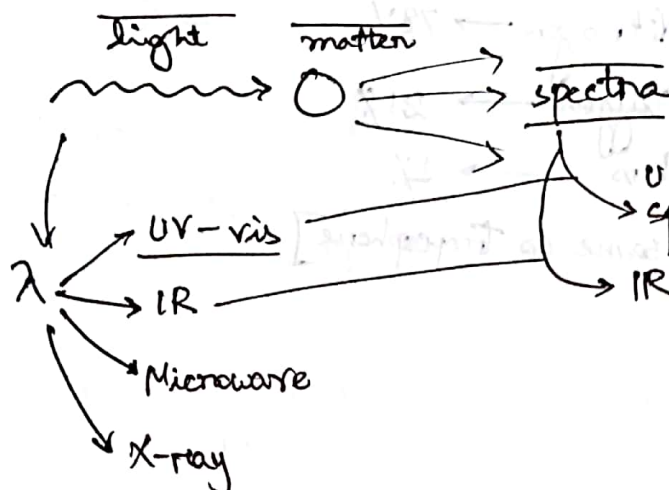
* Hydrologic cycle

Topic Name : CHEM 1207 (MS)

Day : Tuesday

Time : 8.50 Date : 26 / 09 / 2023

Spectrophotometry



spectroscopy: reaction between light and matter

spectra analysis

UV-spectrophotometry: detects electronic transition

IR spectrophotometry: where functional groups or molecules present

* Any light cannot surpass/permeate crystal/matter, they get diffracted and spread spectra

* Spectroscopy: can pass through the liquid solution, so it isn't always diffraction

UV-vis spectrophotometry:

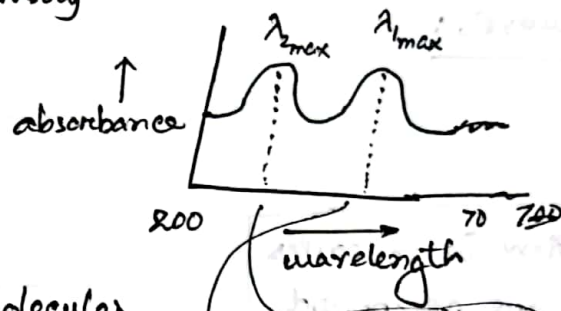
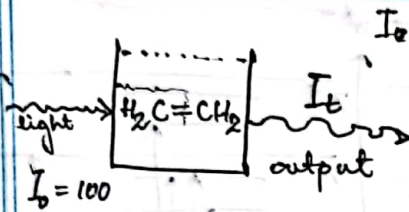
↳ applicable for both atoms and molecules

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Date : / /

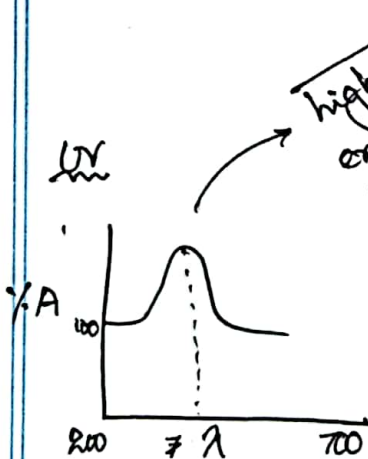
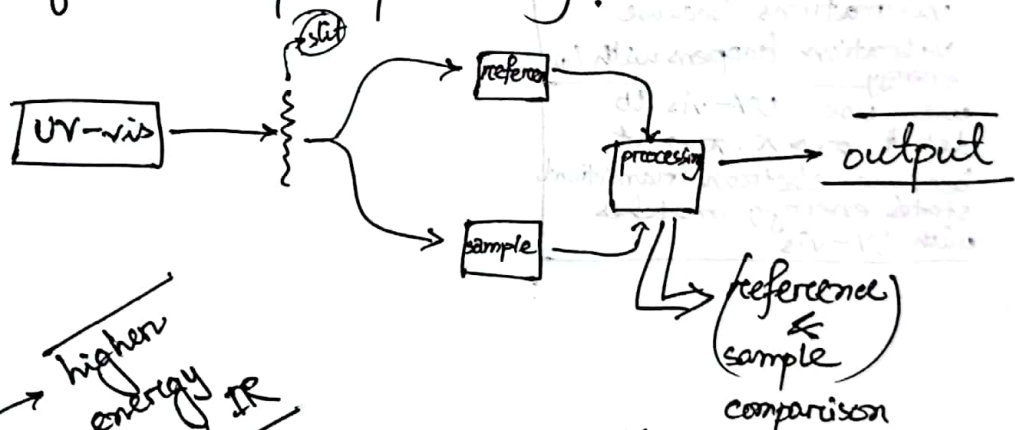


How do we detect molecules using spectrophotometry?

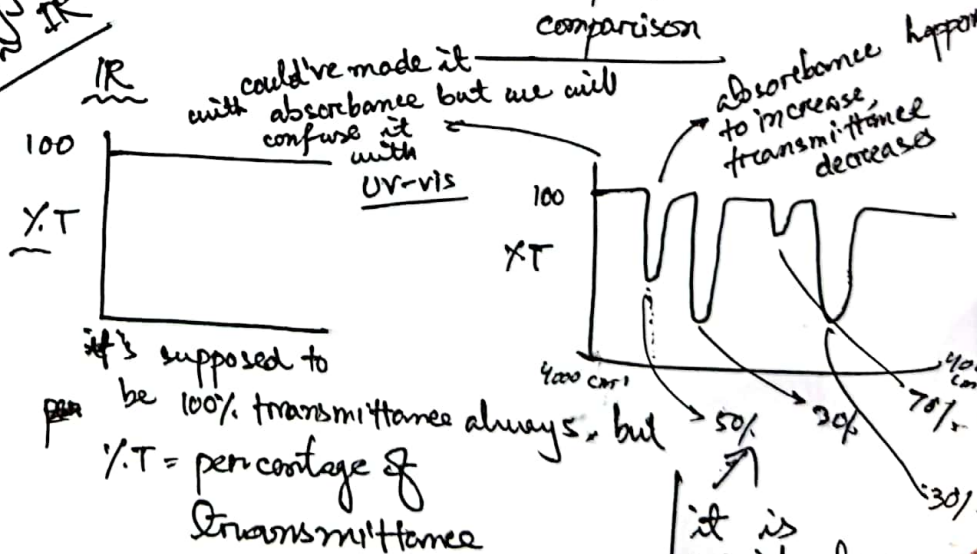
analysis of this graph peaks, searching where these transitions occur, we find out it happens in Ethene. $\text{H}_2\text{C}=\text{CH}_2$

$\pi \rightarrow \pi^*$
 $\pi \rightarrow \sigma^*$

Principles of UV-vis spectrophotometry.



%A = percentage of absorbance



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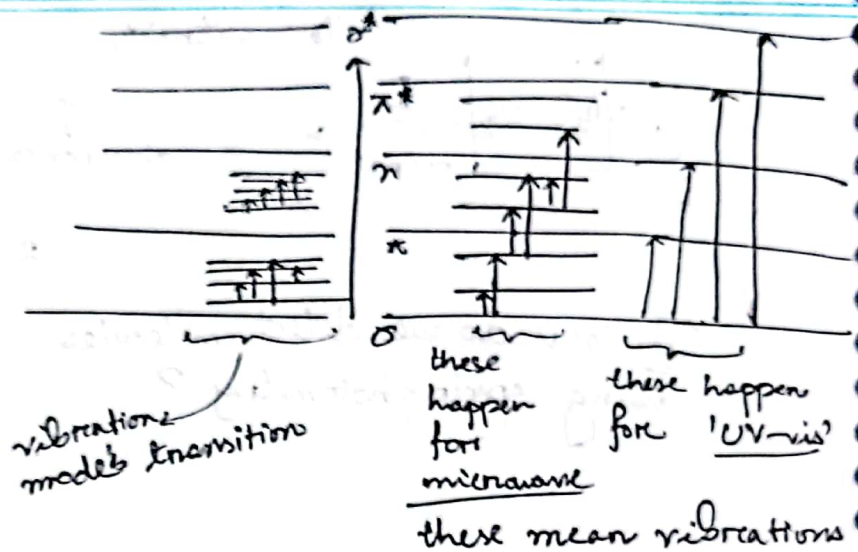
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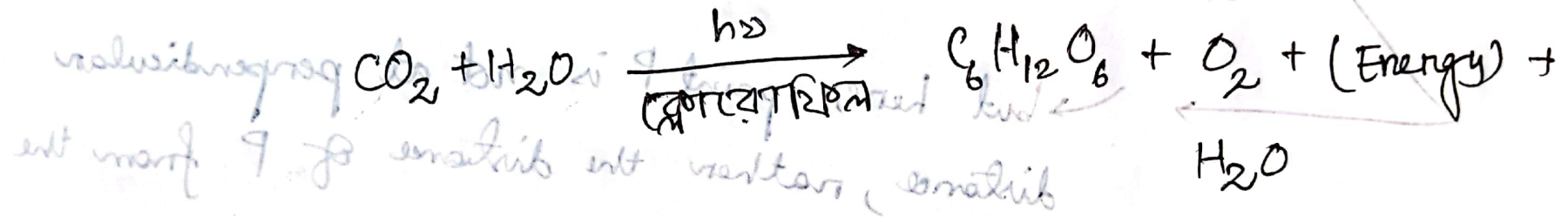
Microwave :

Vibration in molecular bonds are over passed, light doesn't interfere with vibration, but light is used to only detect vibration modes.

we use IR to detect vibrations because vibration happens with low energy

we use UV-vis to detect $\sigma \rightarrow \pi$, $\pi \rightarrow \pi^*$ because electron transition state energy matches with UV-vis





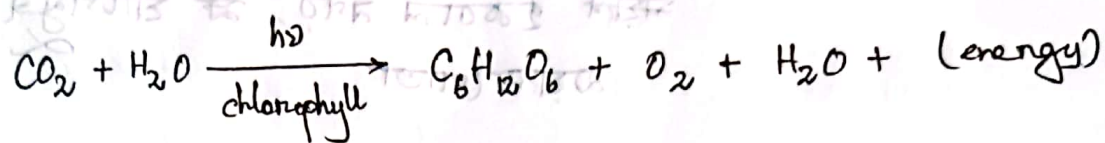
(1) Photosynthesis

(2) Absorption and Emission spectrum

(3) Hydrogen bonding & Titanic accident

(4) Greenhouse effect

(5)

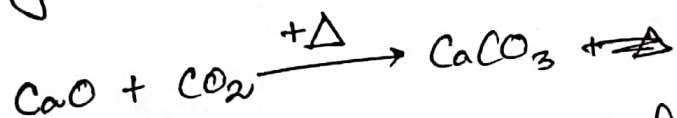


* Photochemical reaction

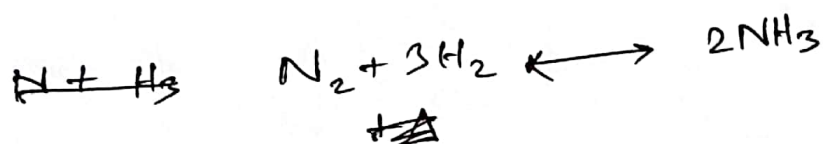
* Thermal reaction

The branch of chemistry which deals with photochemical reactions is called photochemistry.

eg: ~~CaCO₃~~



definition: The reaction which are



Quantum yield:

The number of molecules reacted or formed per photon of light absorbed is termed Quantum Yield.

$$\phi = \frac{\text{number of molecules reacted or formed}}{\text{number of photons absorbed}}$$

$$\phi = \frac{x}{y}$$

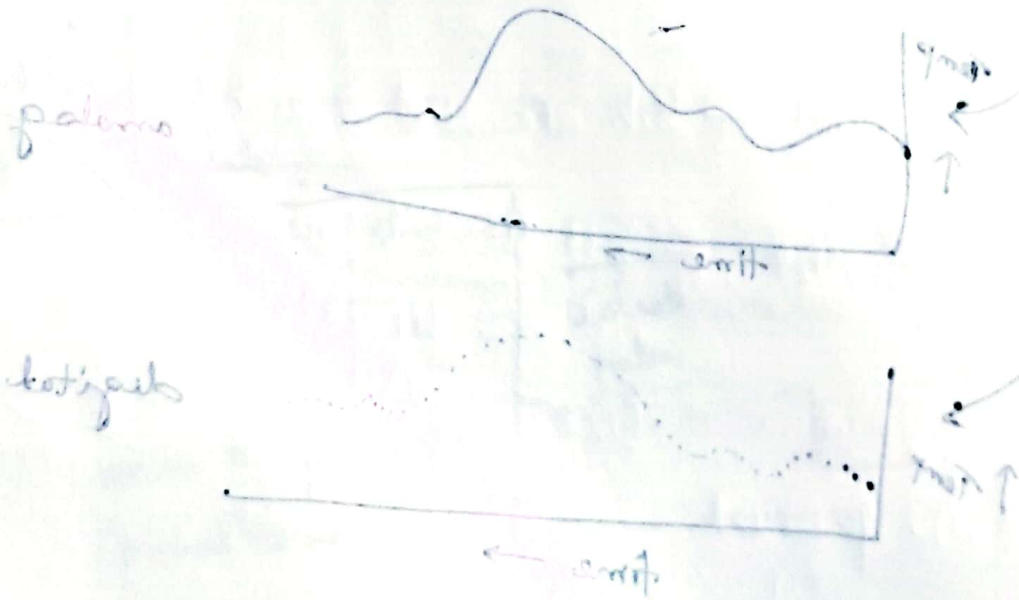
x = number of molecules reacted or formed
 y = number of photons absorbed

When 2 or 2+ are decomposed per photon, $\phi > 1$ and the reaction has a high quantum yield.

When < 1 molecules per photon it is a low quantum yield.

$\phi > 1 \rightarrow$ High Quantum Yield reaction

$\phi < 1 \rightarrow$ Low Quantum Yield reaction



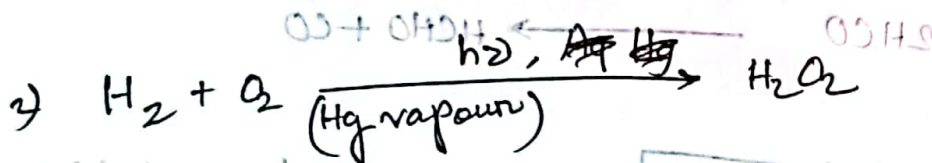
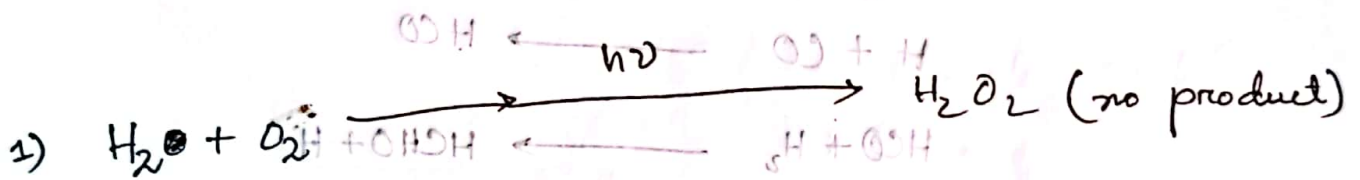
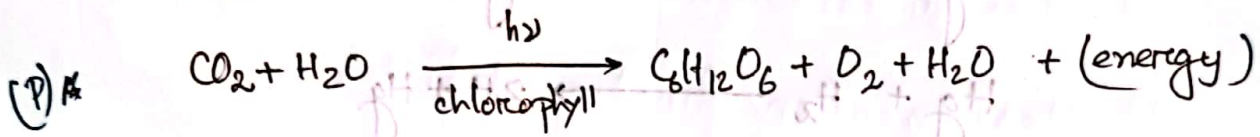
Quantum yield is a measure of the efficiency of a photochemical reaction. It is defined as the number of molecules of a particular product formed per unit volume of the reaction mixture per unit time per unit intensity of incident light.

* ଆପଣଙ୍କ ସହ CLASS DISCUSSION, A4 size paper ଏ ମିଧ୍ୟ
ଆମେ ଶୁଣିବୁ ନେକ୍ସ୍ଟ କ୍ଲାସ୍. ASSIGNMENT

Photochemical Reaction:

Reaction:

Photosynthesis



→ Photo sensitised reaction (p-1053)

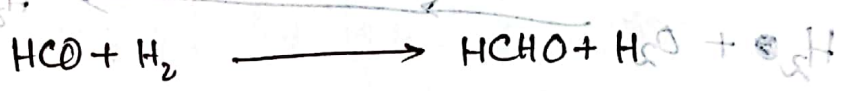
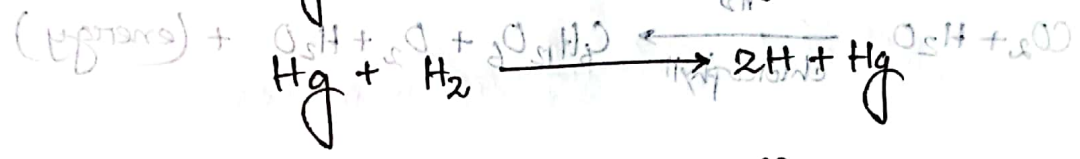
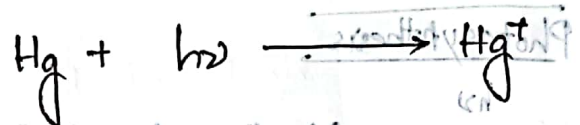
(P) and (2) are very similar

↳ (P) takes the water & $h\nu$ directly and produces product
(2) couldn't, needed Hg vapour to be excited and then releases thermal energy to produce H_2O_2 product.

17.07.21

TEST 1: organic acids PA & HORMONAL DISORDERS
 THYMINE. also look at other

Formaldehyde production:



Quantum efficiency: $\phi > 1$ high quantum yield reaction

Primary reaction:

~~secondary~~ Photochemical reaction

Secondary reaction:

Thermal reaction

examples of hoggy HQY reaction:



reason of HQY: They're using the leftover released energy

(1) class declaration

(2) class member function definition

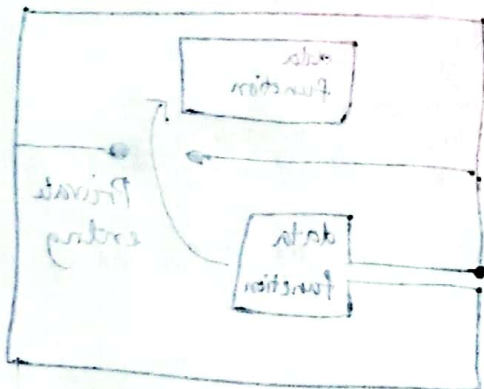
class scope

storing

variable → data
function → code

public

variable
function



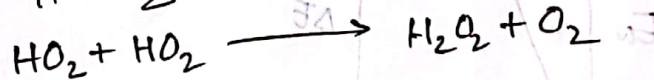
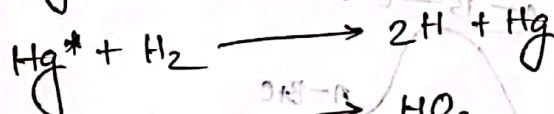
member function characteristics:

• several different classes can use the same function name. The identifier label will resolve their scope.
• to access the private data of class

• we can call all member functions directly without tab notation

* Photosensitizers are a kind of catalysts. eg: Hg vapour

* Photosensitized reaction

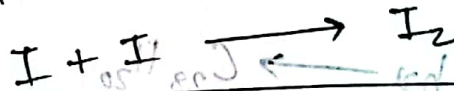
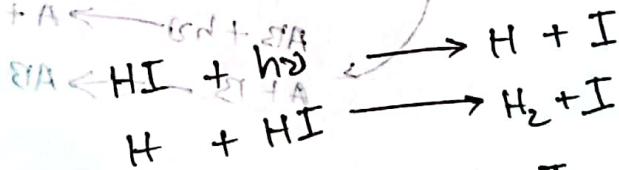


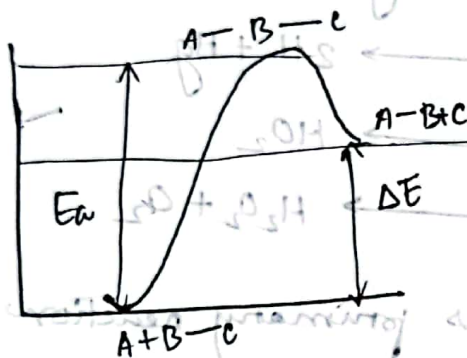
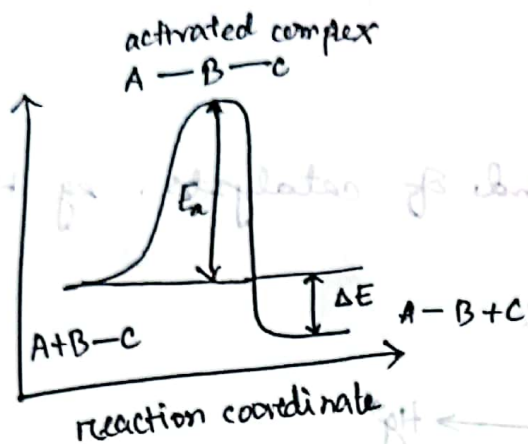
* Photosynthesis reaction is primary reaction

causes of H₂O₂:

one photon absorbed in a primary reaction ...

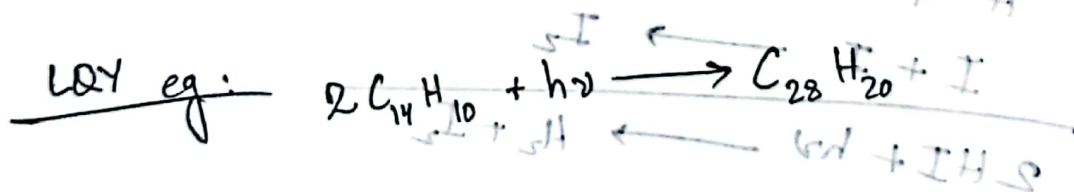
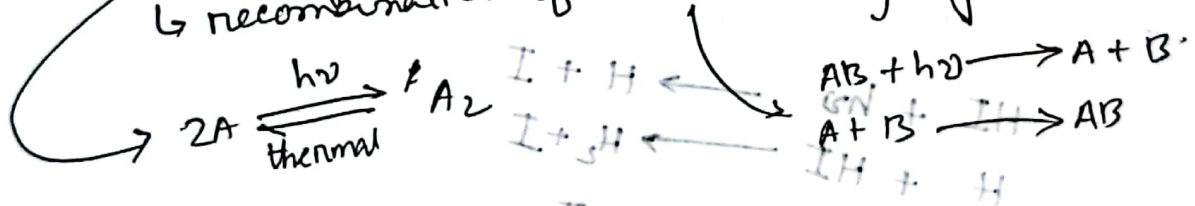
... occurs in the presence of a sensitizer ...





causes of LQY:

- ↳ deactivation of reacting molecules
- ↳ occurrence of reverse of primary reaction.
- ↳ recombination of dissociated fragments.



Laws of photochemistry :

Grothius - Draper law :

Stark - Einstein law : Photochemical equivalence law

Grothuss-Draper law:

only that much light is active which is absorbed.

Stark-Einstein:

per photon in a photochemical reaction.

→ US Scientists researched out these products for 30 years:

* Lipitor → used for heart diseases

* 5-Fluoro Uracil → Fluorine creates polarity, for this

* Geoloda → for cancer

Fluorescence:

Certain molecules (or atoms) when exposed to light radiation of short wavelength (high frequency) emit light of longer wavelength, this pt is called FLUORESCENCE.

eg:

(a) a solution of quinine sulphate on exposure to visible light, exhibits blue fluorescence.

(b) a solution of chlorophyll in ether show blood red fluorescence.

* Phosphorescence!

When a substance absorbs radiation of high frequency and emits light even after the incident radiation is cut off, this is called PHOSPHORESCENCE.

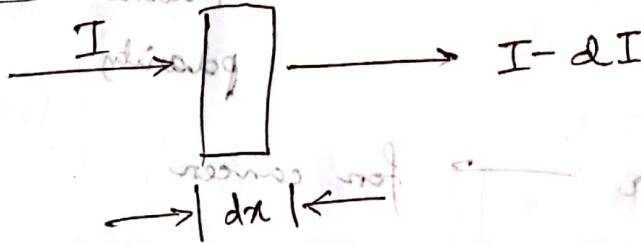
* Chemiluminescence:

The emission of light as a result of chemical reaction is called CHEMILUMINESCENCE.

* Lambert-Beer's Law:

$$\ln\left(\frac{I}{I_0}\right) = -bx$$

$$\ln\left(\frac{I}{I_0}\right) = -\epsilon Cx$$



Fluorescence is the emission of light by a substance that has absorbed light or other electromagnetic radiation. It is a form of luminescence. In most cases, the emitted light has a longer wavelength (lower frequency) than the absorbed radiation. This is called FLUORESCENCE.

(a) A solution of a substance can be excited by light, and it will emit fluorescence. (b) A solution of a substance can be excited by other means, such as heat, and it will emit fluorescence.

$$E = h\nu$$
$$= \frac{hc}{\lambda}$$

$$N = 6.02 \times 10^{23}$$

$$100N = 6.02 \times 10^{23}$$

of Thermopile

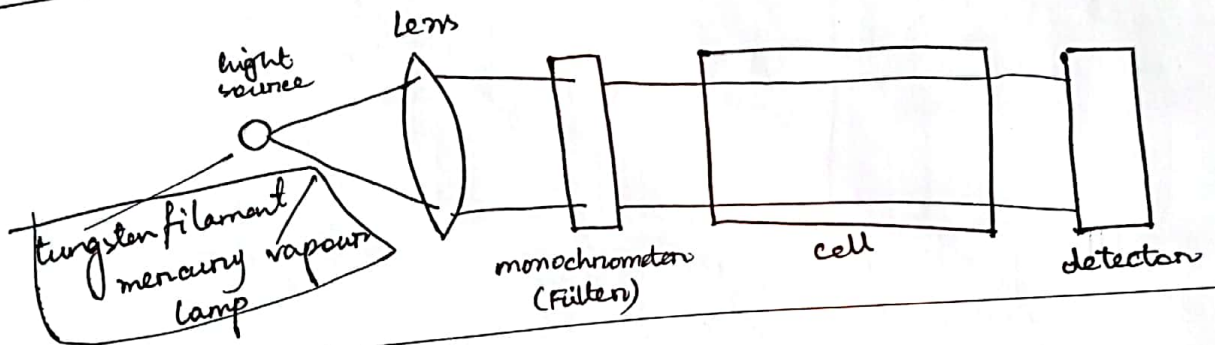
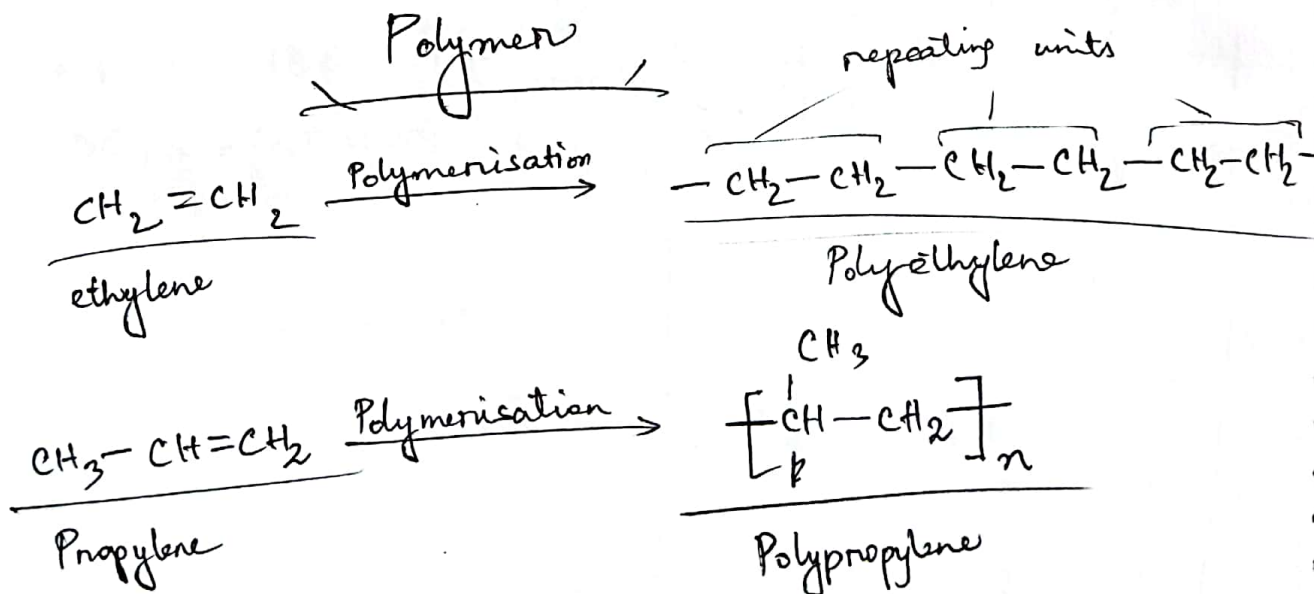


Fig. Schematic diagram of the apparatus used for measurement of light intensity



Types of Polymer:

↳ conducting polymer

↳ non-conducting polymer

⇒ mechanism of addition polymerisation:

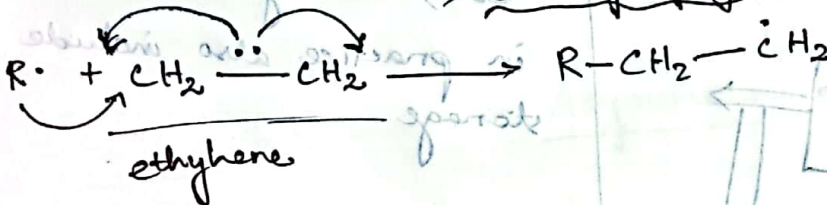
↳ Free radical polymerisation

↳ Chain initiation:

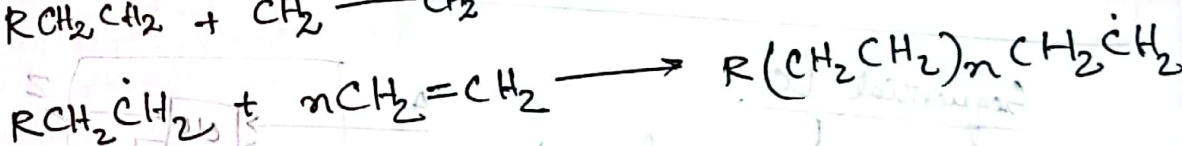
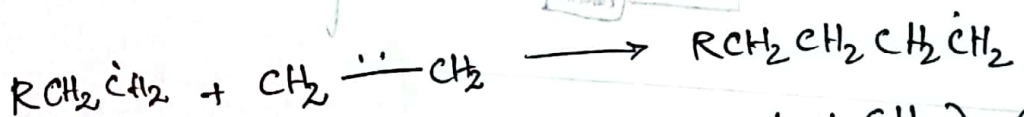


↳ This bond breaks

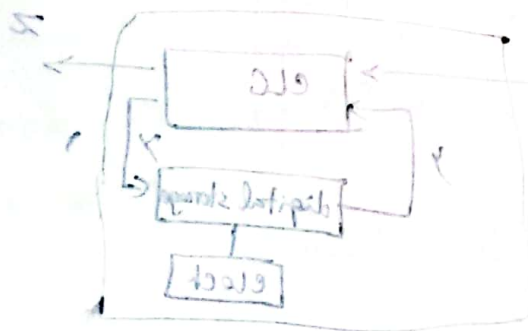
↳ Chain propagation:

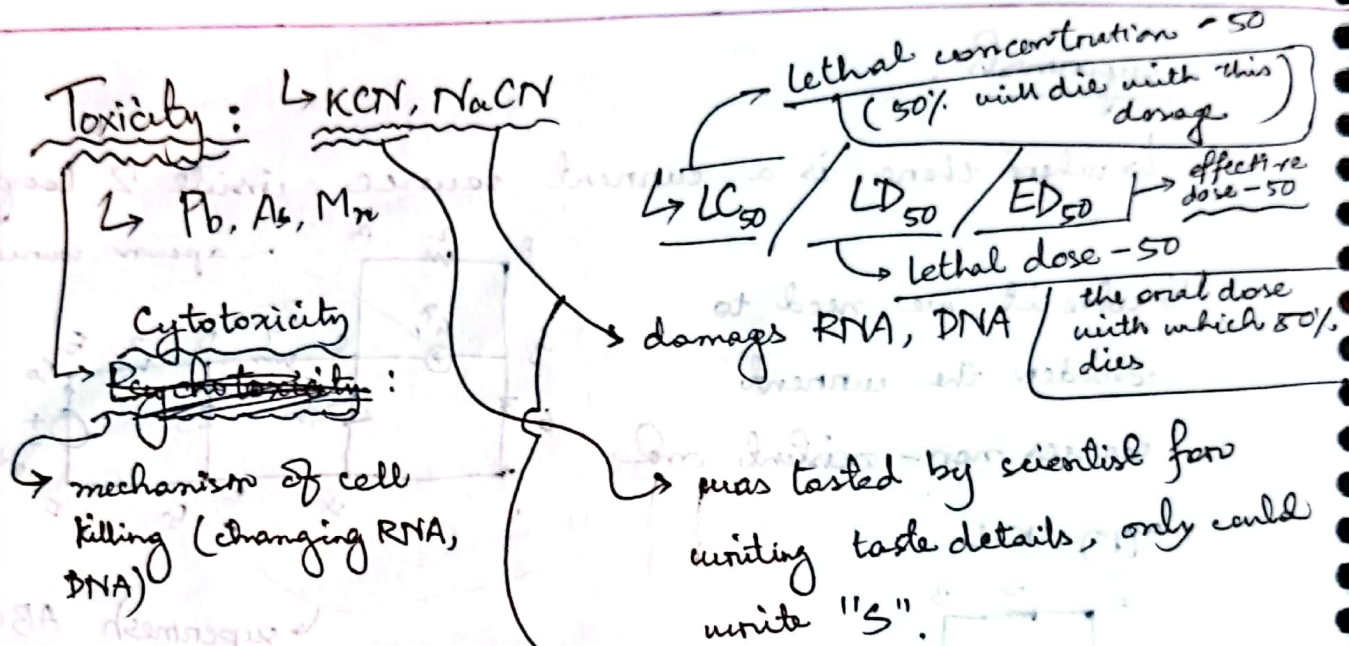


This free radical can attack another molecule and so on.



↳ Chain termin





- $\rightarrow$ How KCN, NaCN, Pb, As, Mn work their toxicity?
- $\rightarrow$ Nature of their toxicity, how they damage cells, RNA and DNA structure.

$\rightarrow$ search youtube

ASSIGNMENT

15-20 pgs

$\rightarrow$ Subacute toxicity:

$\rightarrow$ Chronic toxicity

Paracetamol LD_{50}

$\rightarrow$ 2000 mg/kg

Ciprocin LD_{50}

$\rightarrow$ >2000 mg/kg

Polymer:

high molecular weight compounds

→ ~~conducting~~

→ non-conducting → polythene, polypropylene, PET,
nylon, (Thermoplastics)

→ ~~will melt on heat~~
will soften → melt → burn on heat

Thermoplastic:

A thermoplastic polymer is one which softens on heating and becomes rigid again on cooling. This is because there are weak attractive forces between the long polymer molecules and these are readily disrupted on heating. Most addition polymers are and some condensation polymers are thermoplastic.

eg: Nylon, Polythene, Polystyrene

Thermosetting plastic:

A thermosetting polymer is one which becomes hard on heating. It cannot be softened by heating.

Bakelite → scientist Bakelite discovered this and got Nobel Prize

Polymers of this type are often prepared in 2 stages.

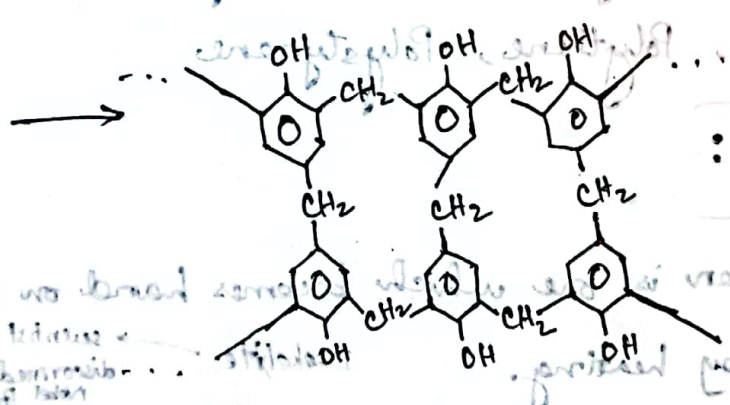
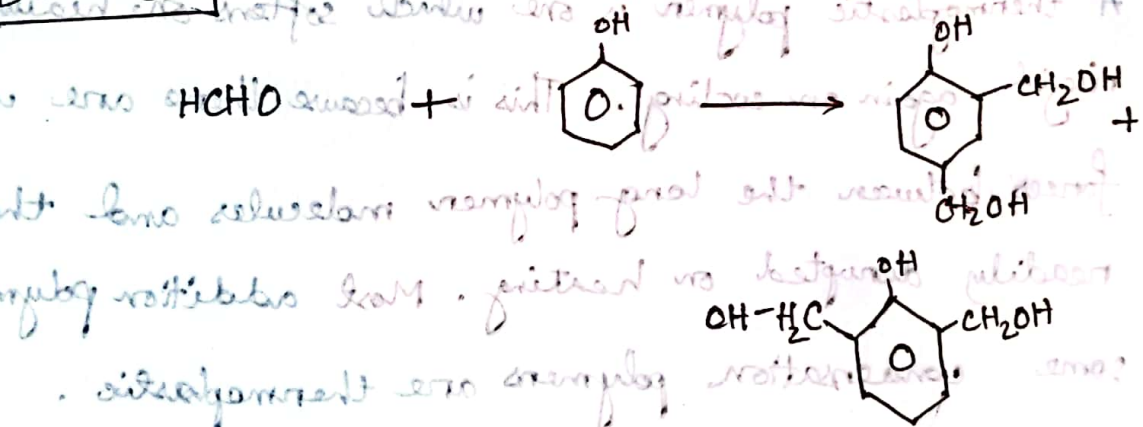
→ Formation of long chain molecules which are capable of further reaction with each other. These intermediate polymers usually flow and can be placed in moulds.

Colouring is often added at this stage.

↳ The application of heat which causes a reaction to occur between the chains, thus producing a complex network polymer.

eg: Phenolic resins, Epoxy resins, Polyurethanes, Alkyd resins

Bakelite:



Bakelite

→ widely used for making molded products such as handles on electrical and cooking utensils; electric plates and switches, appliances; and as a bonding adhesive in plywood

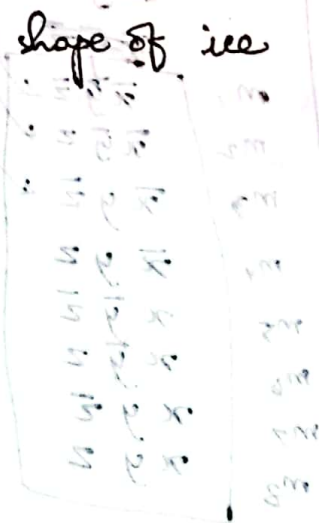
Why does Nylon not tear so easily?

→ Hydrogen bonding

intermolecular
intramolecular

this is created due to electronegativity difference

Hexagonal shape of ice

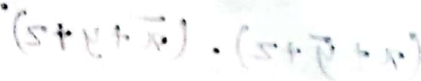


method is also...
molecule of water...
hydrogen bonds...



0 quadrupole
polar
(positive end)

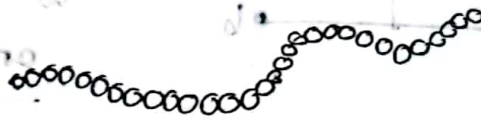
normal phase



(quadrupole) 1 quadrupole
which is polar

(1) Types of polymers :

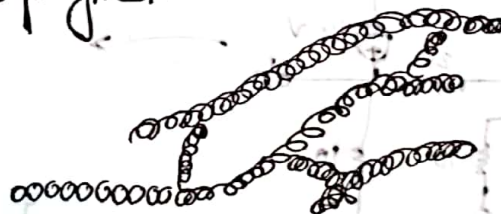
* Linear polymer



* Branched polymer



* Cross linked polymer



(2) Addition and condensation polymers :

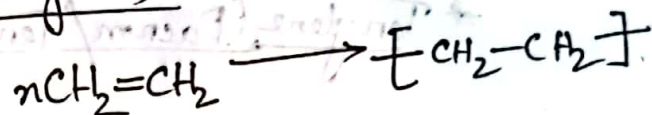
→ only 1 monomer
multiple of that
1 monomer gets
polymerized

* Barack Obama
* Jack Ma
* Akijuddin Sheikh
* Sufi Mohammad
Mizanur Rahman

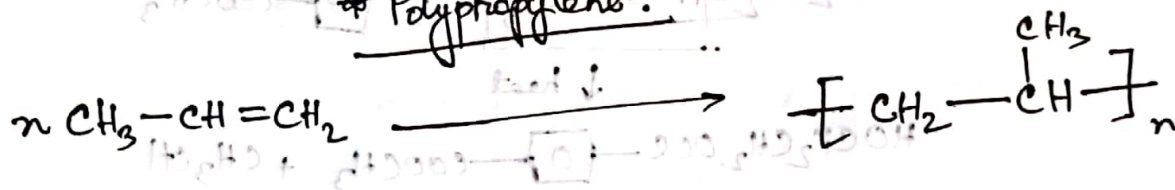
→ ASSIGNMENT on
their life story

* Addition reactions :

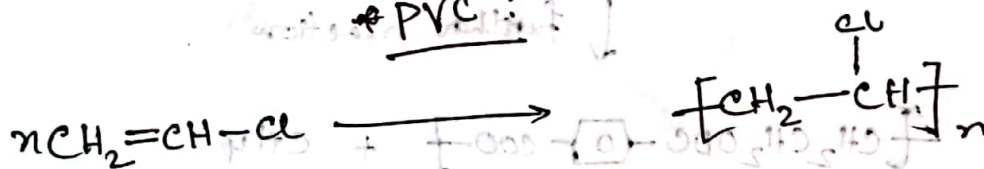
* Polyethylene (Polythene):



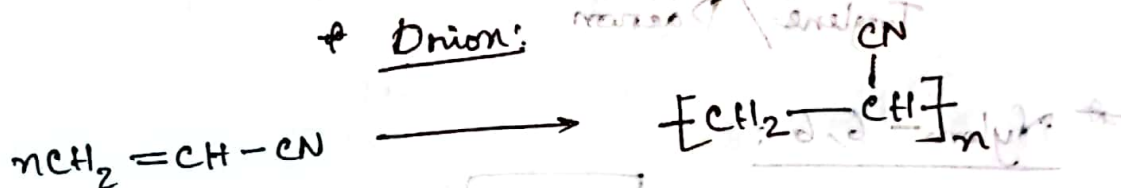
* Polypropylene:



* PVC:



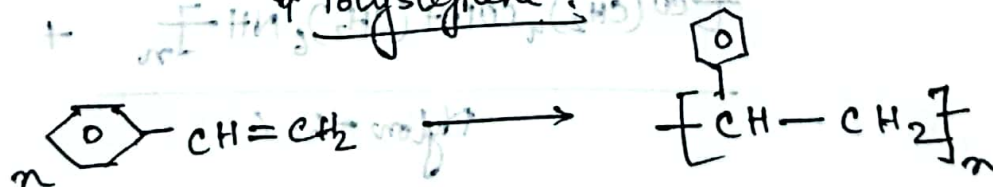
* Orlon:



* Teflon:

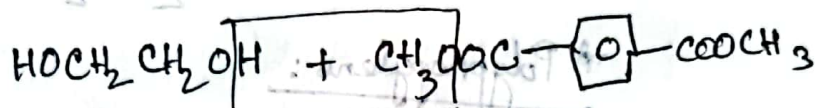


* Polystyrene:

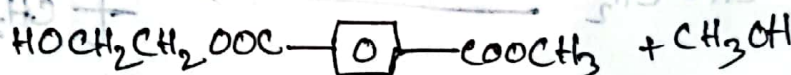


* Condensation polymer:

* Terylene (Dacron/Tenon):



↓ heat

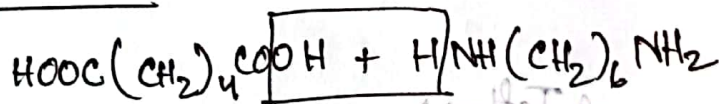


↓ further reaction

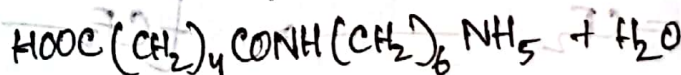


Terylene / Dacron

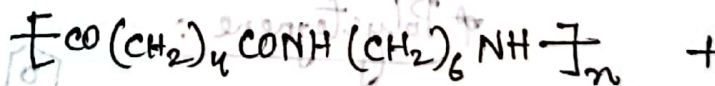
* Nylon-6,6:



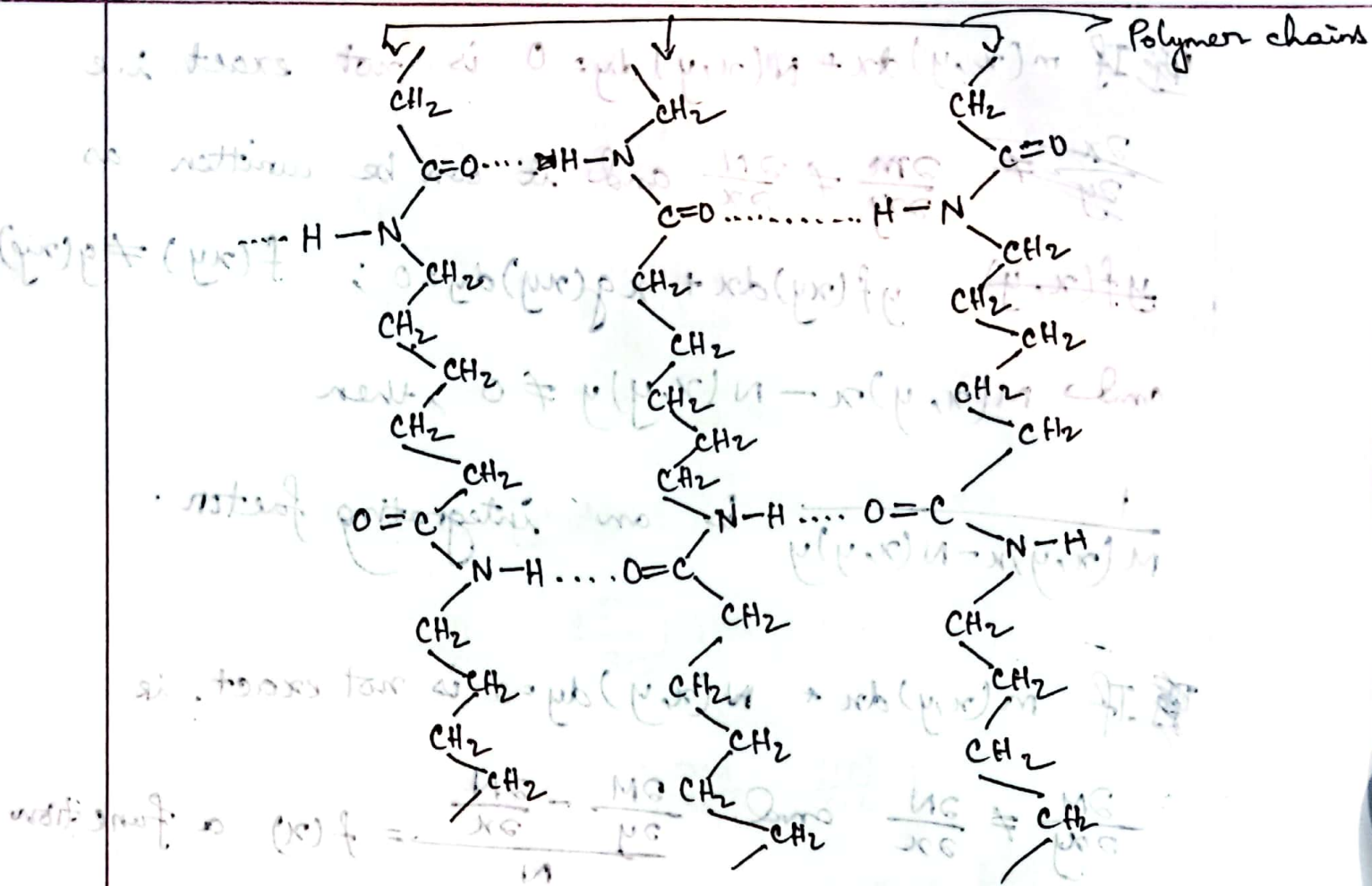
↓ heat



↓ further reaction



Nylon-6,6



①
$$0 = p \cdot b \left(\frac{v_{ps} + x}{p} \right)^{\frac{1}{p}} + x \cdot b \left(\frac{v_{ps}}{p} + \frac{v_{ps}}{p} + p \right)$$

$$\left(\frac{v_{ps} + x}{p} \right)^{\frac{1}{p}} = \frac{v_{ps}}{p} + \frac{v_{ps}}{p} + p$$

$$\left(\frac{v_{ps} + x}{p} \right)^{\frac{1}{p}} = \frac{v_{ps}}{p}$$

$$\frac{v_{ps} + x}{p} = \left(\frac{v_{ps}}{p} \right)^p$$

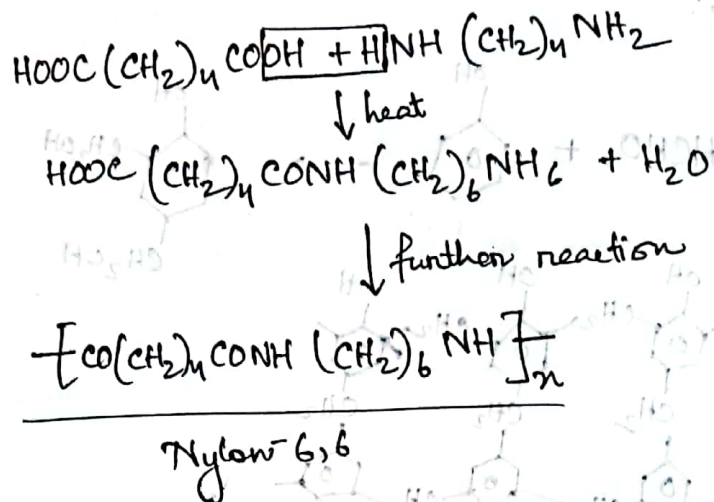
$$v_{ps} + x = p \left(\frac{v_{ps}}{p} \right)^p$$

$$v_{ps} + x = \frac{p \cdot v_{ps}^p}{p^p}$$

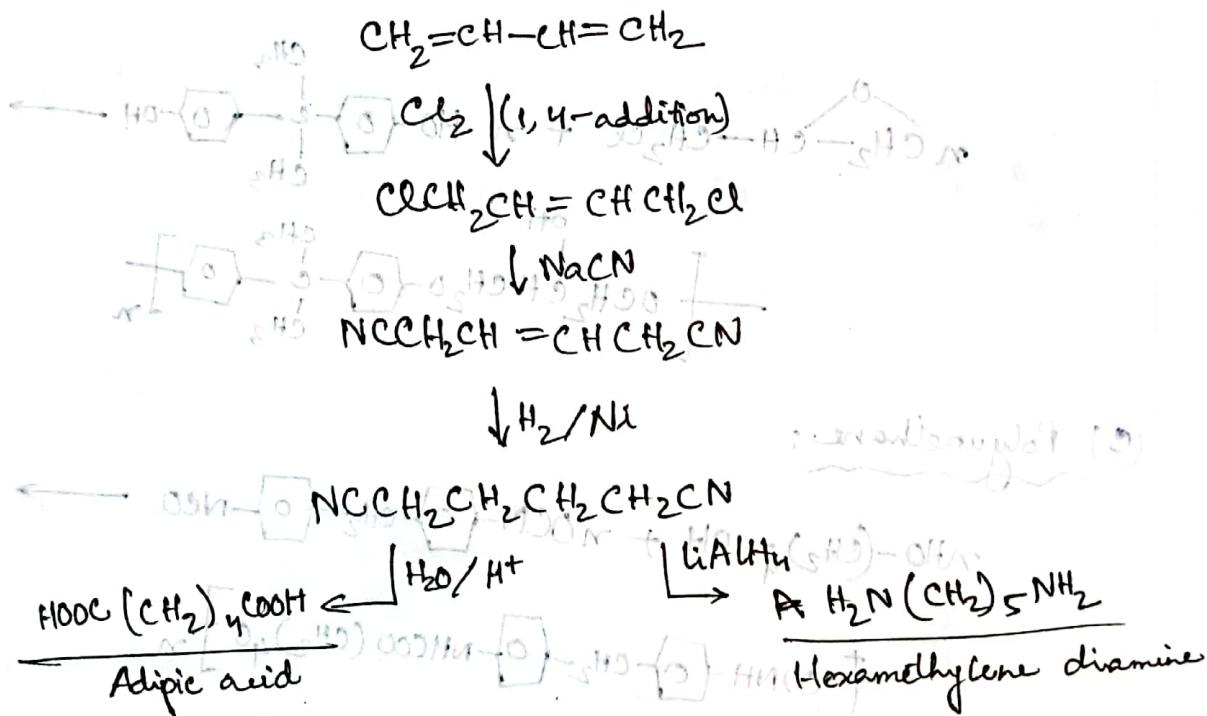
$$\frac{v_{ps}}{p} = \frac{v_{ps}^p}{p^p}$$

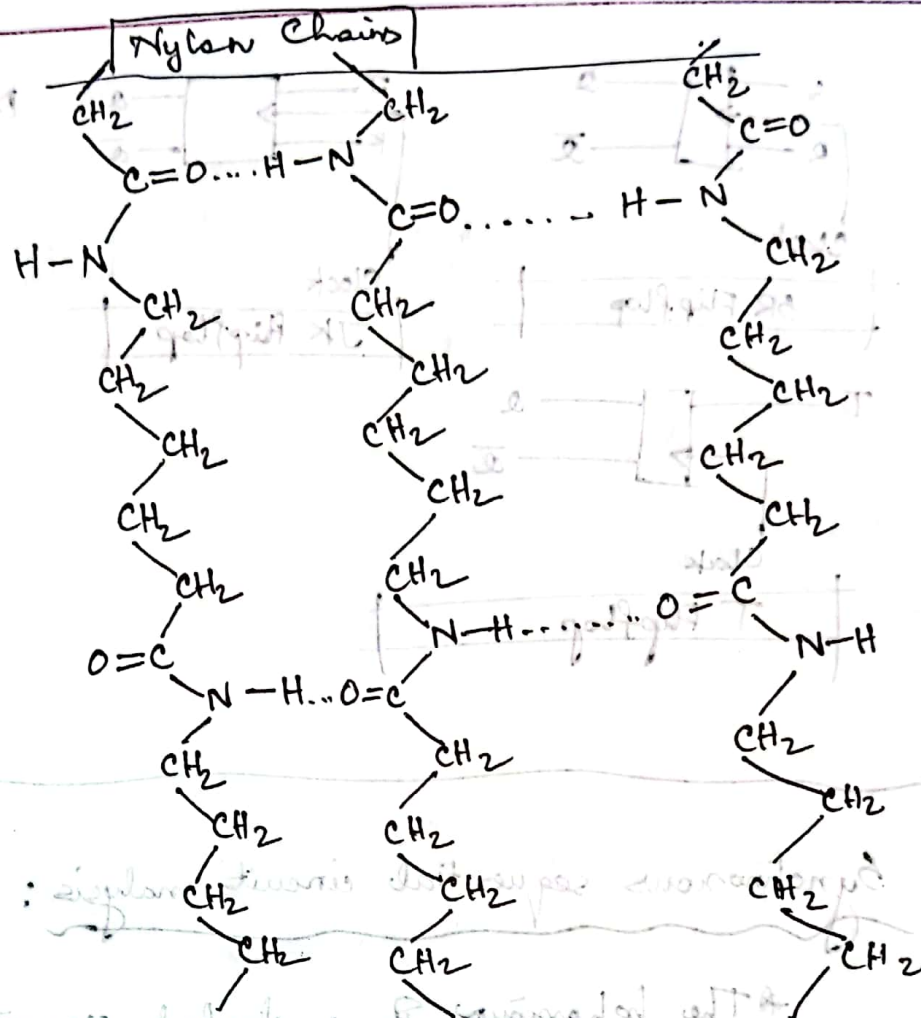
Addition polymers:↳ Polyethylene (Polythene):↳ Polypropylene:↳ Polyvinyl Chloride (PVC):↳ Orlon (Acrilan):↳ Teflon (PTFE or Fluon):↳ Polystyrene:

Nylon 6,6:

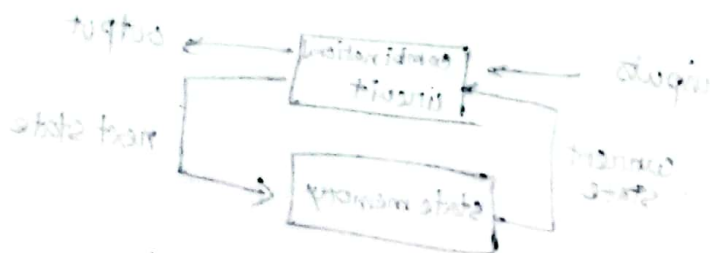


* both starting materials can be prepared from 1,3-butadiene

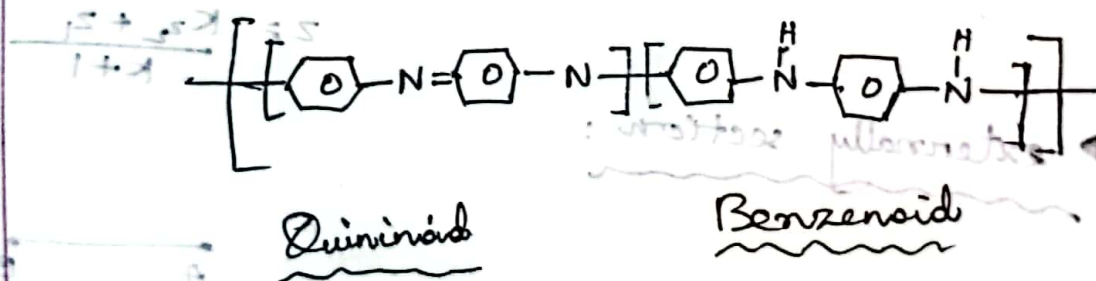
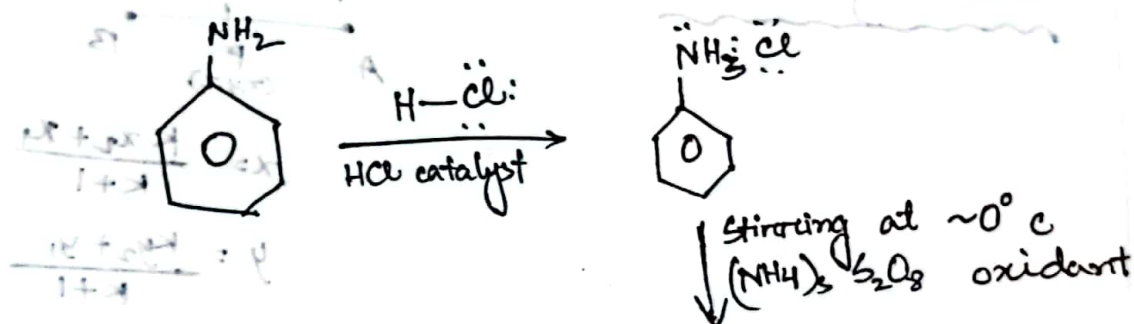
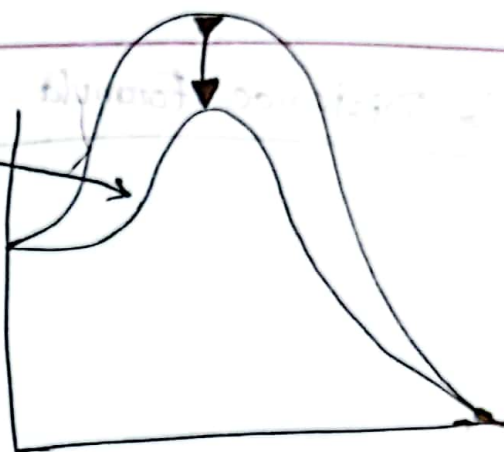




NYLON - 6,6



reducing
activation energy



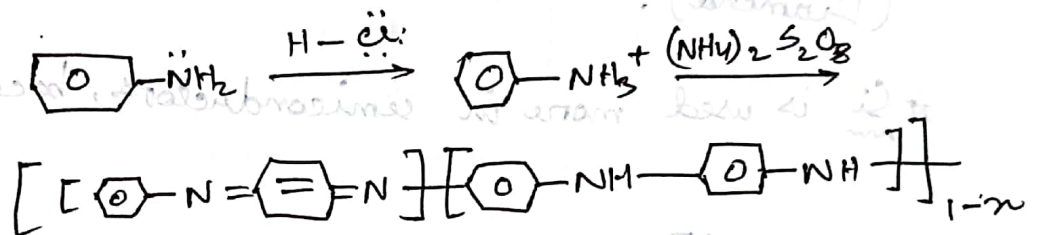
Topic Name : CHEM 1207 (MHM)

Day : Monday

Time : 11.30 Date : 28/08 / 23

Superconductors $\rightarrow$ polyaniline

* synthesis of polyaniline :



* XRD is used for measuring size and shape of crystal.

Degree of Polymerization :

→ the number of monomeric units in a macromolecule or polymer or oligomer molecule

molecular weight of polymer, M_n

molecular weight of monomer, M_0

$$\therefore DP_n = \frac{M_n}{M_0} = \text{degree of polymerization}$$

$$\text{Teflon's } DP_n = \frac{M_n}{M_0} = \frac{120000}{100} = 1200$$

viscosity method :

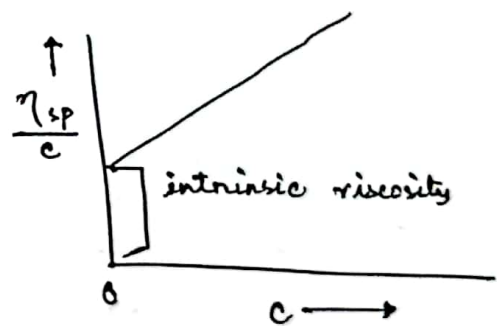
* relative viscosity: $\eta_{rc} = \frac{\eta}{\eta_0}$ — (i)

* specific viscosity: $\eta_{sp} = \eta_{rc} - 1$ — (ii)

* intrinsic viscosity: $[\eta] = \lim_{c \rightarrow 0} \left(\frac{\eta_{sp}}{c} \right)$

$= kM^a$ → M may be calculated from a determination of the value of intrinsic viscosity

viscosity means the property of an element



Conductance :
 → metallic → how it conducts?
 → polymeric → how it conducts?

Topic Name : _____

Day : _____

Time : _____

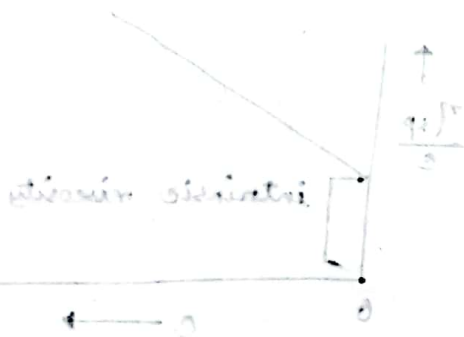
Date : / /

Electron sea model : electron configurations are same, but valence bond electrons float, like in sea.

Mechanism of polymeric conduction :

☐ Compare the conductivity of metallic and polymeric

ASSIGNMENT



① $\frac{1}{T}$: constant, decrease
② $\frac{1}{T}$: decrease, activation energy
③ $\frac{1}{T}$: decrease, activation energy

$\left(\frac{1}{T}\right)_{\text{min}} = \left(\frac{1}{T}\right)_{\text{max}}$: intrinsic conductivity

$\sigma_{\text{min}} =$

conductivity means the property of an element

Conductance : $\leftarrow$ metallic $\leftarrow$? decreases to what $\leftarrow$ how it conducts?
 $\leftarrow$ polymeric $\leftarrow$? decreases to what $\leftarrow$ how it conducts?

Topic Name : CHEM 1207 (MHM)

Day : Monday

Time : 11.30 Date : 18 / 09 / 23

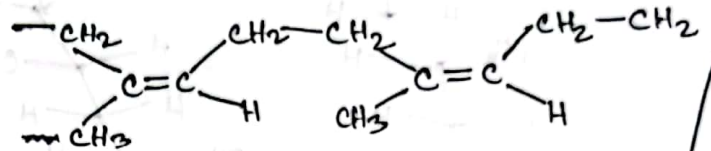
Latex : 60% rubber

* natural rubber

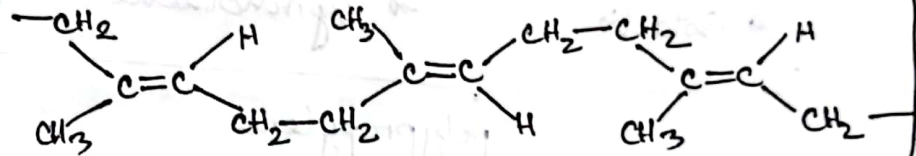
* synthetic rubber

polyisoprene

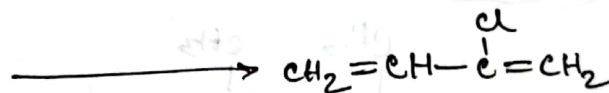
cis configuration of natural rubber



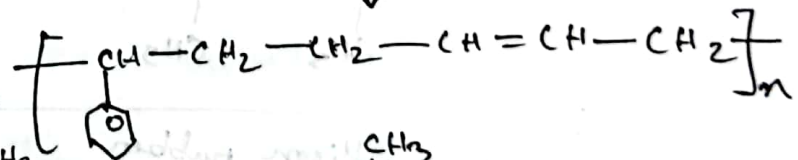
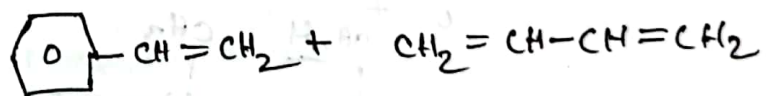
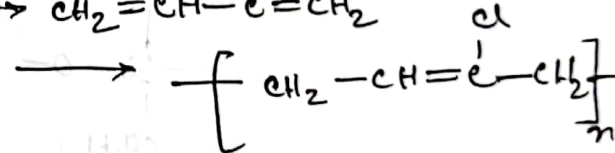
trans configuration of natural rubber



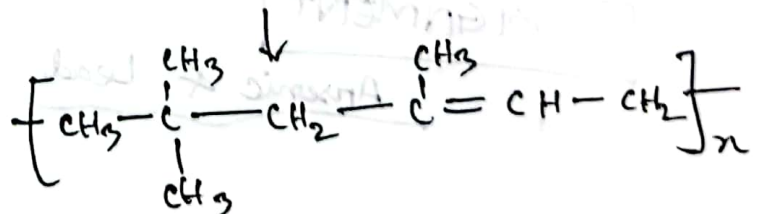
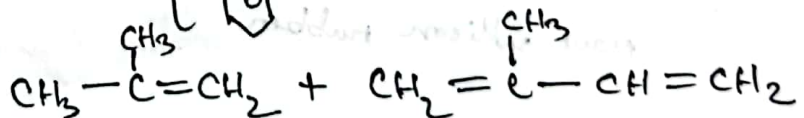
* Neoprene (polychloroprene)



* Styrene-Butadiene rubber



* Butyl rubber:

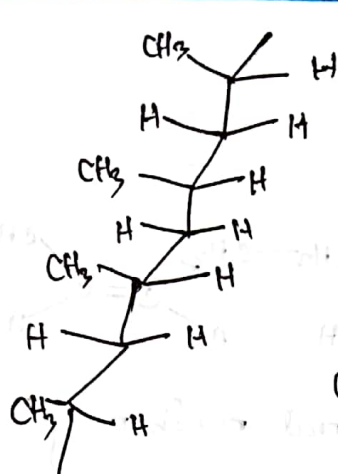


Topic Name : _____

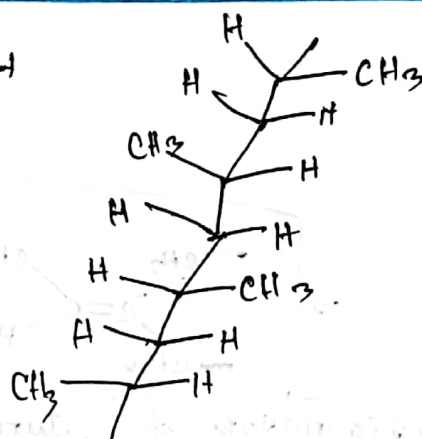
Day : _____

Time : _____

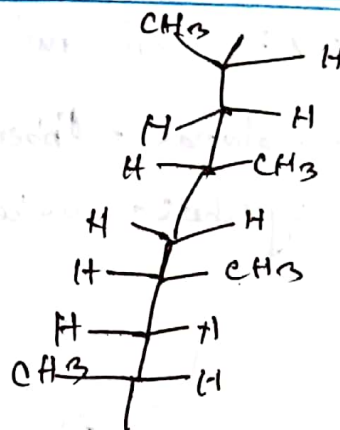
Date : / /



isotactic



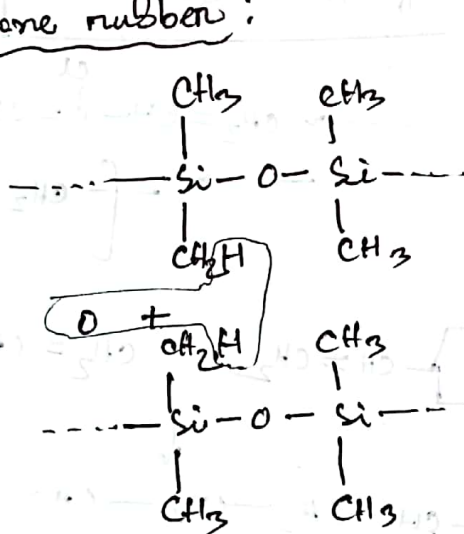
syndiotactic



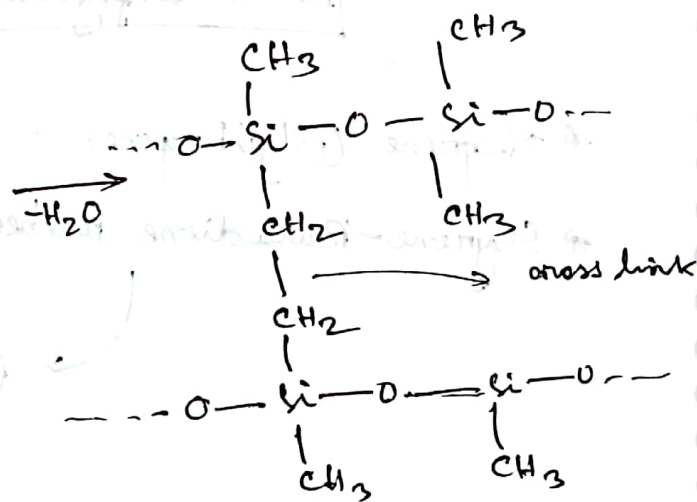
atactic

polypropylene

Silicone rubber :



raw silicone rubber



vulcanized silicone rubber

ASSIGNMENT

Arsenic & Lead